

Supporting Information

Energetically more stable singlet cyclopentane-1,3-diyl diradical with π -single bonding character than the corresponding σ -single bonded compound

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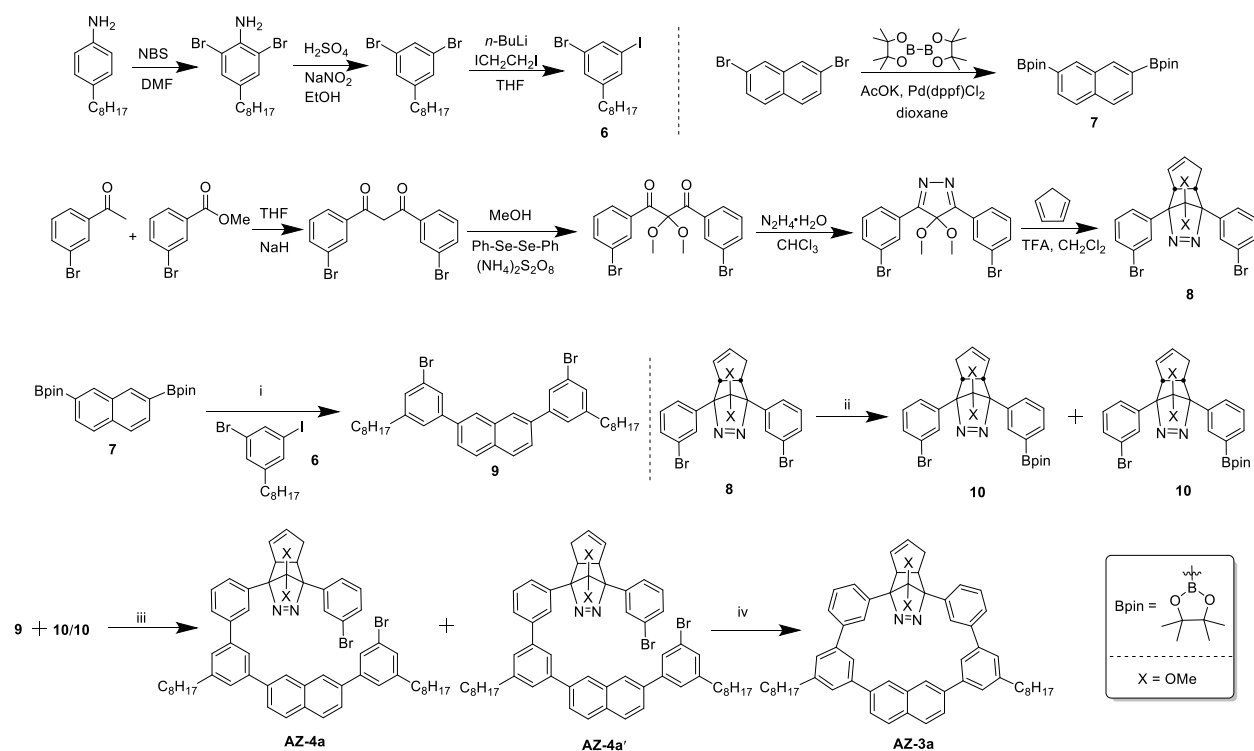
Materials and Methods

General Information

All reagents were purchased from commercial suppliers (TCI, Wako or Sigma Aldrich) and used without further purification. Nuclear magnetic resonance (NMR) spectra (^1H : 400 MHz and ^{13}C : 100 MHz) were measured on a Bruker Ascend 400 spectrometer at 298 K. All chemical shifts (δ) are denoted in ppm and coupling constants (J) in Hz. The resonance multiplicities were represented by abbreviation (s = singlet, d = double, t = triplet, m = multiplet). High resolution mass spectra (HRMS) were obtained on a Thermo Fisher Scientific LTQ Orbitrap XL or JMS-T100GCV AccuTOF GCv. Recycling chromatographic separation in GPC were performed by LC-9210, Japan Analytical Industry Co., Ltd. with chloroform as mobile phase.

The sub-microsecond transient absorption (TA) measurement under Ar or N_2 atmosphere: The sample solution was placed in a four-sided 1 cm quartz cuvette, covered with a septum, and Ar or N_2 bubbling was conducted for 20 min. Temperature sensor (CHINO corporation, model DB1010B000–G0A) was inserted into the cuvette through the septum, and then Ar or N_2 balloon was attached. The measurement temperature was controlled by using CoolSpeK (Unisok Co., Ltd., model USP-203) and temperature value of the temperature sensor was used for measurement and analysis. The TA measurement under vacuum condition: The sample solution was placed in a four-sided 1 cm quartz cuvette and air was removed by vacuum line 3~4 times. The temperature was controlled by CoolSpeK (Unisok Co., Ltd., model USP-203). The decay signals at 580 nm upon 355-nm excitation using a 355 nm yttrium aluminum garnet (YAG) laser (4.5 mJ/pulse, 5 ns pulse width) was measured at each temperature. The lifetime values (τ) were determined by single exponential fitting of first-order decay signals.

Quantum chemical computations in gas phase were carried out with Gaussian 16 [Revision. B.01] suite of programs. Charge, spin multiplicity, imaginary frequencies, energies [in Hartree] and atomic coordinates [in Angstroms] of optimized structures at (R,U)B3LYP/6-31G(d) level of theory are listed in Data S1. Local minima and transition state structures were confirmed by vibrational frequency analysis. The transition states were further confirmed by intrinsic reaction coordinate (IRC) calculation with the alternative local quadratic approximation (LQA) approach (Figure S11). The bonding orbital and anti-bonding orbital and their electron occupation numbers in π -**3b** were calculated at CASSCF wave functions includes 2 active electrons and 2 active orbitals, namely, CASSCF[2,2] with the 6-31G[d] basis set.



Scheme S1. General procedure for synthesizing the azoalkanes.

Synthesis of AZ-3a

Compounds **6**, **7** and **8** were synthesized according to the literatures and their characterization including NMR spectroscopies (¹H and ¹³C) and HRMS have been reported¹.

(i) A round flask with a condenser was charged with **7** (1.12 g, 2.94 mmol, 1.0 equiv.), **6** (2.58 g, 6.52 mmol, 2.2 equiv.), Ag₂CO₃ (3.26 g, 11.82 mmol, 4.0 equiv.) and dry THF (55 mL). The reaction flask was replaced with nitrogen atmosphere after freeze pump thaw cycling (four times). Then tetrakis(triphenyl-phosphine)palladium(0) (0.34 g, 0.29 mmol, 0.1 equiv.) was added into the flask under nitrogen atmosphere. The reaction mixture was refluxed at 65 °C for 20 h. After cooling to room temperature, the reaction mixture was filtered through a pad of celite and concentrated. The crude mixture was purified by silica gel column (hexane only) to afford **9** (0.55 g, 0.83 mmol, 28 %) as colorless liquid. ¹H NMR (400 MHz, CDCl₃): δ □ 8.07 (s, 2H), 7.94 (d, *J* □ 8.38 Hz, 2H), 7.73–7.68 (m, 4H), 7.46 (s, 2H), 7.35 (s, 2H), 2.67 (t, *J* □ 7.71 Hz, 4H), 1.68 (m, 4H), 1.38–1.25 (m, 20H), 0.88 (t, *J* □ 7.01 Hz, 6H). ¹³C NMR (100 MHz, CDCl₃): δ □ 145.68, 142.80, 137.74, 133.77, 132.17, 130.39, 128.32, 127.70, 126.27, 126.22, 125.61, 122.95, 35.92, 32.02, 31.46, 29.59, 29.46, 29.40, 29.45, 22.83, 14.28. HRMS (APCI) calculated for C₃₈H₄₆Br₂ [M]⁺ □ 660.19608, found 660.19641.

(ii) Under exclusion of light, to a side-arm flask was added **8** (2.52 g, 5.00 mmol, 1.0 equiv.), [1,1'-bis(diphenylphosphino)ferrocene]dichloropalladium(II) (0.20 g, 0.25 mmol, 5% equiv.), bis(pinacolato)diboron (1.27 g, 5.00 mmol, 1.0 equiv.), potassium acetate (1.47 g, 14.98 mmol, 3.0 equiv.) and 1,4-dioxane (24 mL). The reaction mixture was stirred at 70 °C for 20 h under nitrogen atmosphere. After cooling to room temperature, the mixture was filtered through a pad of celite and solvent was removed in vacuo. The residue was purified by short silica gel column chromatography (CH₂Cl₂ only) and gel permeation chromatography (CHCl₃ only) to give **10/10'**

(1.00 g, 2.36 mmol, 47 %) as a yellow solid. $^1\text{H NMR}$ (400 MHz, CDCl_3): δ 8.41 (s, 1H), 8.31 (s, 1H), 8.15 (t, J 1.72 Hz, 1H), 8.07 (t, J 1.72 Hz, 1H), 8.02–7.99 (m, 1H), 7.96–7.92 (m, 2H), 7.87–7.81 (m, 3H), 7.55–7.51 (m, 2H), 7.49–7.43 (m, 2H), 7.37–7.31 (m, 2H), 5.57 (m, 2H), 5.52 (m, 2H), 4.26 (m, 1H), 4.11 (m, 1H), 3.80 (m, 1H), 3.65 (m, 1H), 2.99 (s, 6H), 2.69–2.66 (m, 6H), 2.35 (m, 2H), 2.22 (m, 2H), 1.36 (s, 24H). **HRMS** (ESI) calculated for $\text{C}_{28}\text{H}_{32}\text{N}_2\text{O}_4\text{BBrNa}$ $[\text{M}^+\text{Na}]^+$ 573.15307, found 573.15247.

(iii) In a round bottom flask, **9** (2.29 g, 3.46 mmol, 2.0 equiv.), Na_2CO_3 (0.56 g, 5.28 mmol, 3.1 equiv.) and tetrakis(triphenyl-phosphine)palladium(0) (99.8 mg, 0.09 mmol, 0.05 equiv.) were dissolved in mixture solvents containing EtOH (8.6 mL), toluene (8.6 mL) and H_2O (8.6 mL) under nitrogen atmosphere. On the other hand, **10/10'** was dissolved in dry toluene (8.6 mL) with another round bottom flask. These two flasks were charged with nitrogen atmosphere after freeze pump thaw cycling (four times). The solution of **10/10'** was added to the solution of **9** slowly and stirred for 18 h at 65°C in absence of light under nitrogen atmosphere. Afterwards, the resulting mixture was filtered through a pad of celite and extracted with CH_2Cl_2 . The crude was purified by silica gel column (hexane / CH_2Cl_2 1: 1) and slight yellow solid mixture **AZ-4a/AZ-4a'** was obtained (0.48 g, 0.48 mmol, 28 %). $^1\text{H NMR}$ (400 MHz, CDCl_3): δ 8.34 (s, 1H), 8.24 (s, 1H), 8.18 (s, 3H), 8.11–8.07 (m, 3H), 7.98–7.92 (m, 6H), 7.88–7.80 (m, 6H), 7.76–7.68 (m, 6H), 7.60–7.46 (m, 10H), 7.37–7.32 (m, 4H), 5.63–5.49 (m, 4H), 4.30–4.24 (m, 1H), 4.18–4.12 (m, 1H), 3.84–3.77 (m, 1H), 3.74–3.66 (m, 1H), 3.04–3.02 (m, 6H), 2.80 (t, J 7.92 Hz, 4H), 2.74 (s, 6H), 2.68 (t, J 7.92 Hz, 4H), 2.46–2.19 (m, 4H), 1.81–1.62 (m, 8H), 1.35–1.29 (m, 40 H), 0.88 (m, 12H). **HRMS** (ESI) calculated for $\text{C}_{60}\text{H}_{66}\text{N}_2\text{O}_2\text{Br}_2\text{Na}$ $[\text{M}^+\text{Na}]^+$ 1027.33833, found 1027.33850.

(iv) 1,5-cyclooctadiene (0.09 mL, 0.72 mmol, 4.3 equiv.) and 2,2'-bipyridyl (0.11 g, 0.72 mmol, 4.3 equiv.) were dissolved in dry-toluene (8.6 mL) and dry-DMF (8.6 mL) under nitrogen atmosphere. The reaction flask was replaced with argon atmosphere after freeze pump thaw cycling (four times). $\text{Ni}(\text{cod})_2$ (0.22 g, 0.80 mmol, 4.7 equiv.) was added and replacement of argon atmosphere was conducted. The mixture was stirred at room temperature for 30 min. On the other hand, under nitrogen atmosphere, **AZ-4a/AZ-4a'** (0.17 g, 0.17 mmol, 1.0 equiv.) was dissolved in dry-toluene (34.5 mL). The reaction flask was replaced with argon atmosphere after freeze pump thaw cycling (four times). The solution of **AZ-4a/AZ-4a'** was added to the Ni-catalyst solution dropwise for 22 min at 70°C and the mixture was stirred at 70°C for more 2.3 h, then cooled down to room temperature. The reaction mixture was added to saturated NH_4Cl aq. and stirred for 30 min at room temperature. The organic layer was extracted with EtOAc two times. After filtration and concentration, the crude mixture was obtained. Subsequently, purify the resulting crude by silica gel column chromatography (hexane / CH_2Cl_2 1: 1) and recrystallization to give **AZ-3a** (13.6 mg, 0.016 mmol) in 9 % yield as white solid. Single crystals of **AZ-3a** suitable for X-ray diffraction analysis were obtained by vapor diffusion of CH_3CN into CHCl_3 solution of **AZ-3a** at 5°C . $^1\text{H NMR}$ (400 MHz, CDCl_3): δ 8.59 (s, 1H), 8.52 (s, 1H), 8.43 (s, 2H), 8.06 (s, 1H), 8.02 (s, 1H), 7.94–7.91 (m, 2H), 7.86–7.80 (m, 4H), 7.67–7.54 (m, 8H), 5.75–5.71 (m, 1H), 5.69–5.65 (m, 1H), 4.49–4.43 (m, 1H), 3.98–3.90 (m, 1H), 2.94 (s, 3H), 2.86 (s, 3H), 2.81 (t, J 7.78 Hz, 4H), 2.60–2.33 (m, 2H), 1.76 (m, 4 H), 1.46–1.24 (m, 20H), 0.89 (t, J 7.02 Hz, 6H). $^{13}\text{C NMR}$ (100 MHz, C_6D_6): δ 144.48, 142.21, 142.16, 141.14, 141.05, 141.01, 139.36, 136.98, 136.92, 135.35, 134.65, 132.52, 132.47, 132.17, 130.77, 130.74, 129.30, 129.07, 128.70, 127.32, 127.09, 126.33, 126.22, 126.16, 125.87, 125.14, 125.06, 124.90, 124.84, 119.98, 95.61, 94.10, 53.50, 52.86, 51.51, 40.21, 36.96, 32.52, 32.40, 32.36, 30.04, 29.92, 29.78, 23.13, 14.37. **HRMS** (ESI) calculated for $\text{C}_{60}\text{H}_{66}\text{N}_2\text{O}_2\text{Na}$ $[\text{M}^+\text{Na}]^+$ 869.50165, found 869.50195.

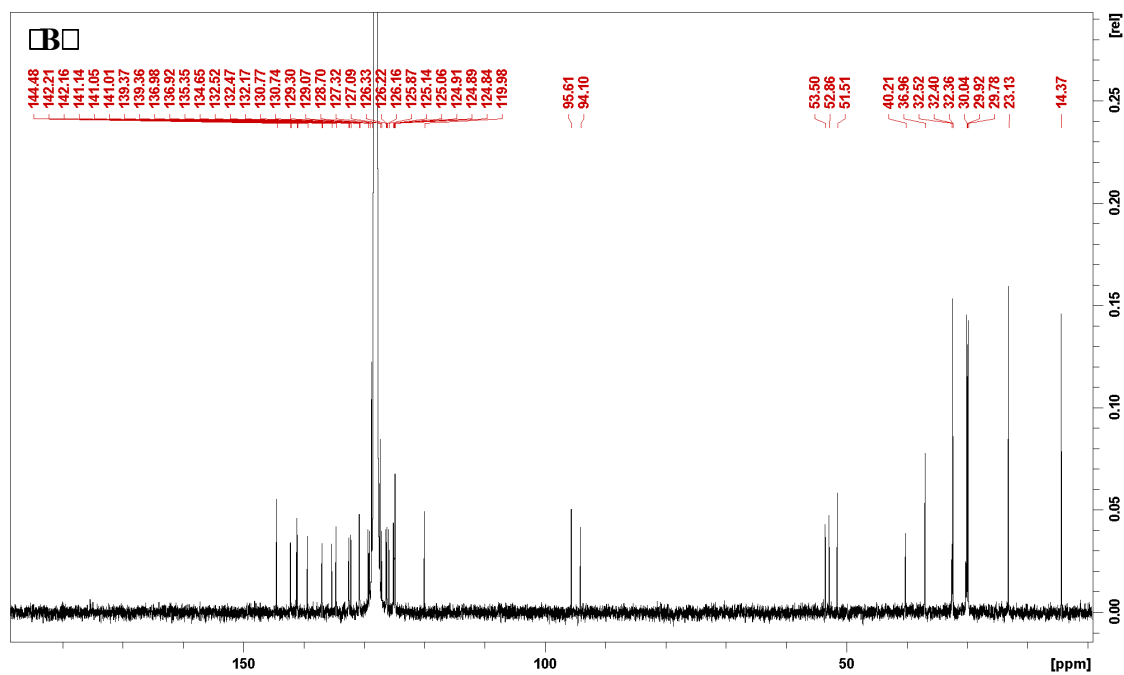
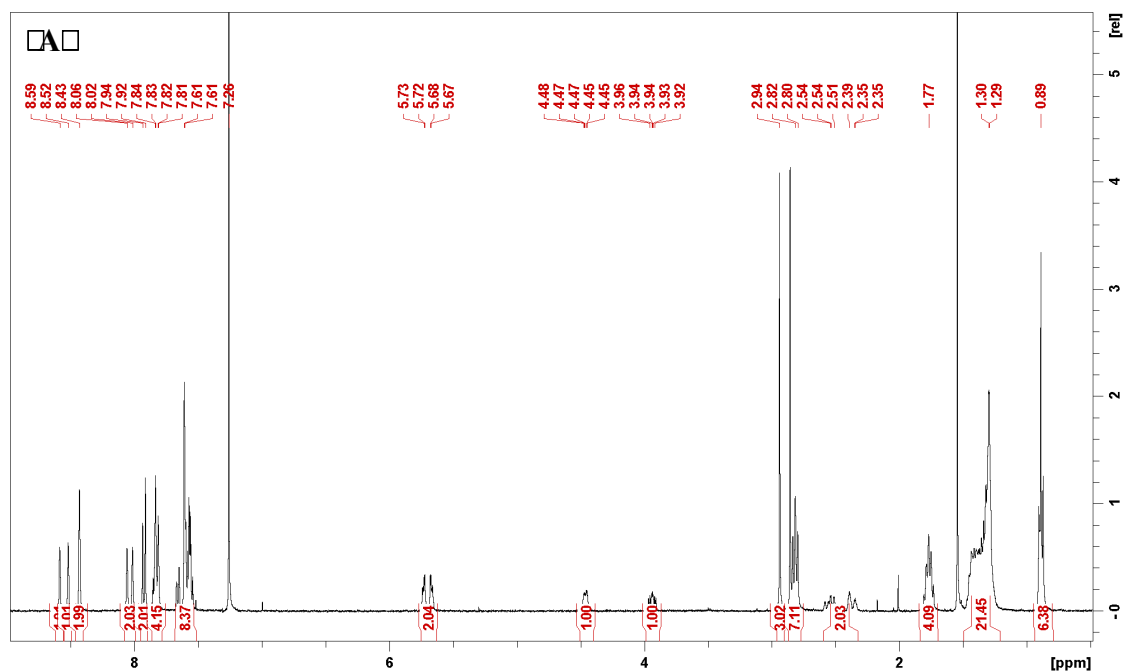


Table S1. Crystal data and structure refinement for AZ-3a.

Bond precision:	C–C □ 0.0031 Å	Wavelength □ 0.71073
Cell:	a □ 24.0668 (17)	b □ 7.8399 (6) c □ 50.719 (4)
	alpha □ 90	beta □ 90 gamma □ 90
Temperature:	173 K	
	Calculated	Reported
Volume	9569.7 (13)	9569.7 (12)
Space group	P b c a	P b c a
Hall group	– P 2ac 2ab	– P 2ac 2ab
Moiety formula	C60 H66 N2 O2	?
Sum formula	C60 H66 N2 O2	C60 H66 N2 O2
Mr	847.15	847.14
Dx, g cm ^{–3}	1.176	1.176
Z	8	8
Mu (mm ^{–1})	0.070	0.070
F000	3648.0	3648.0
F000'	3649.35	
h, k, lmax	27, 9, 58	27, 9, 58
Nref	7631	7624
Tmin, Tmax	0.000, 0.497	0.440, 0.540
Tmin'	0.000	
Correction method □ □ Reported T Limits: Tmin □ 0.440 Tmax □ 0.540		
AbsCorr □ MULTI – SCAN		
Data completeness □ 0.999	Theta max □ 24.160	
R (reflections) □ 0.0462 (5622)	wR2 (reflections) □ 0.1150 (7624)	
S □ 1.008	Npar □ 581	

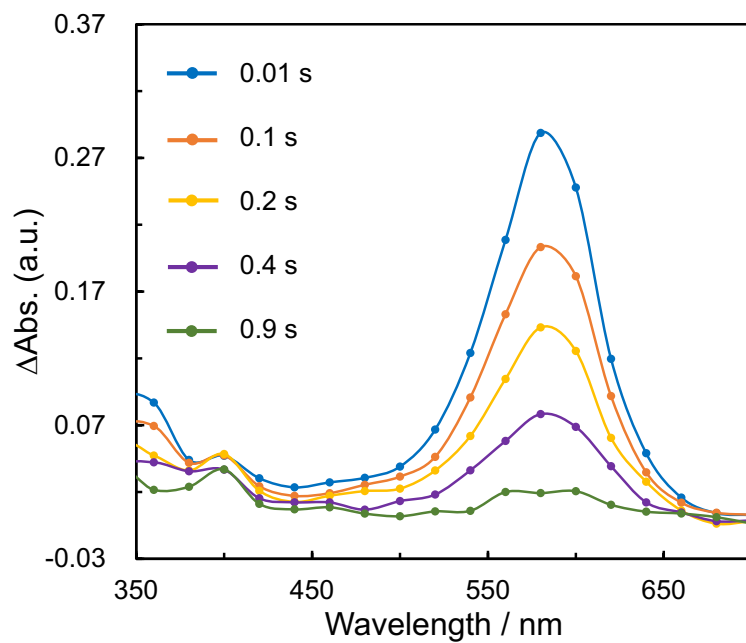


Figure S2. Sub-microsecond transient absorption spectrum during the laser flash photolysis of AZ-3a at 293 K in benzene under vacuum condition.

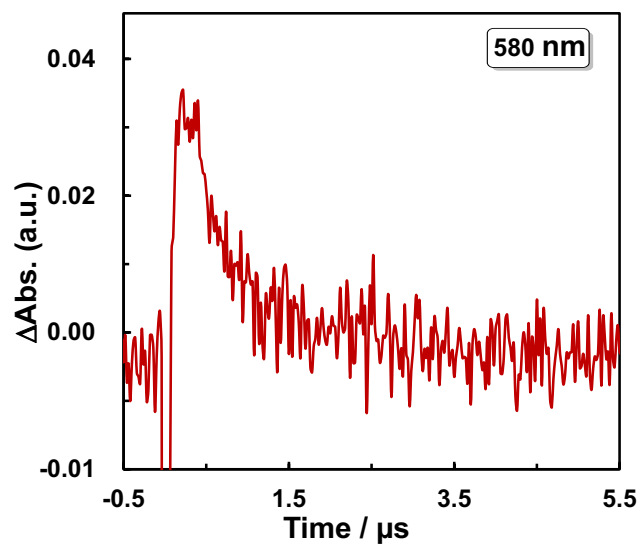


Figure S3. Sub-microsecond transient absorption spectrum during the laser flash photolysis of AZ-4a/AZ-4a' at 293 K in benzene under N₂.

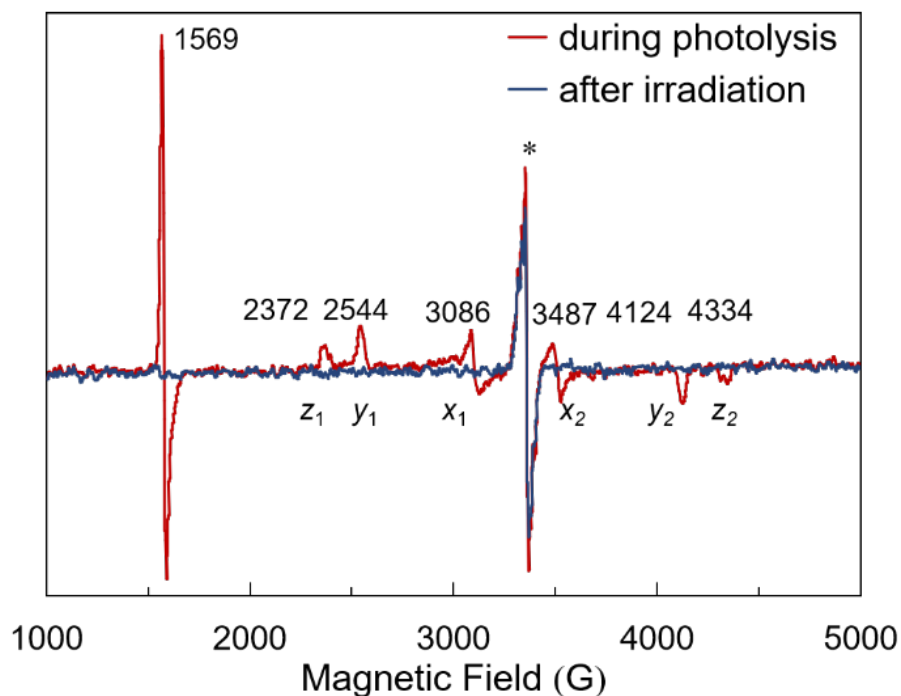


Figure S4. EPR spectra during the photolysis of AZ-**3a** (4.01 mM in a 2-methyltetrahydrofuran (MTHF) matrix at 88 K. Red line, during photolysis and blue line, in the dark after irradiation. The asterisk indicated signals that correspond to impurities in MTHF.

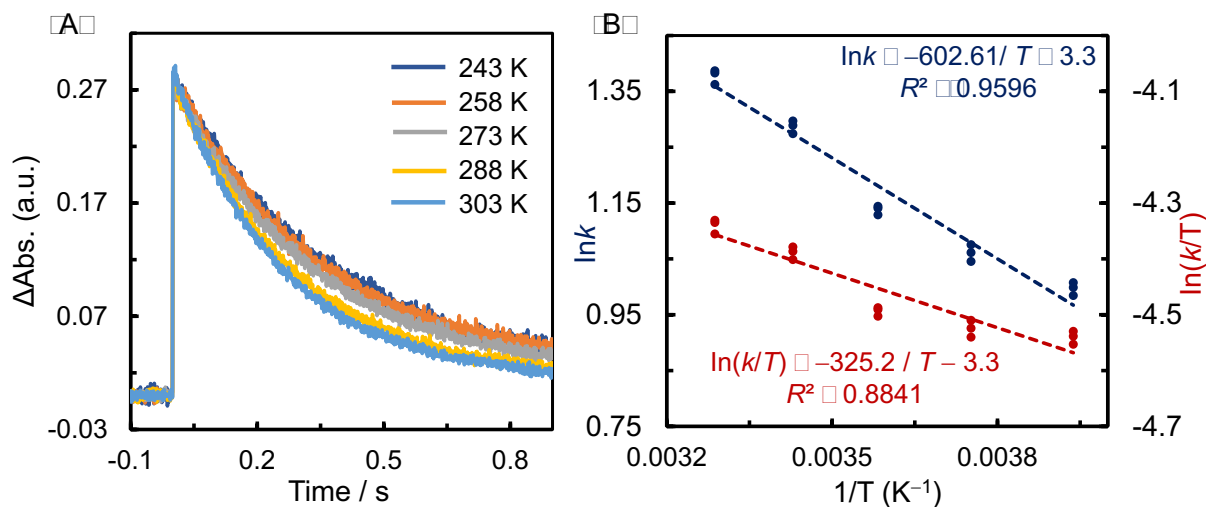


Figure S5. (A) Transient absorption spectrum for the fall process of π -**3a** in toluene under vacuum condition. (B) Arrhenius plots (left y axis, blue line) and Eyring plots (right y axis, red line) for the fall process of π -**3a** in toluene under vacuum condition from 243 K to 303 K.

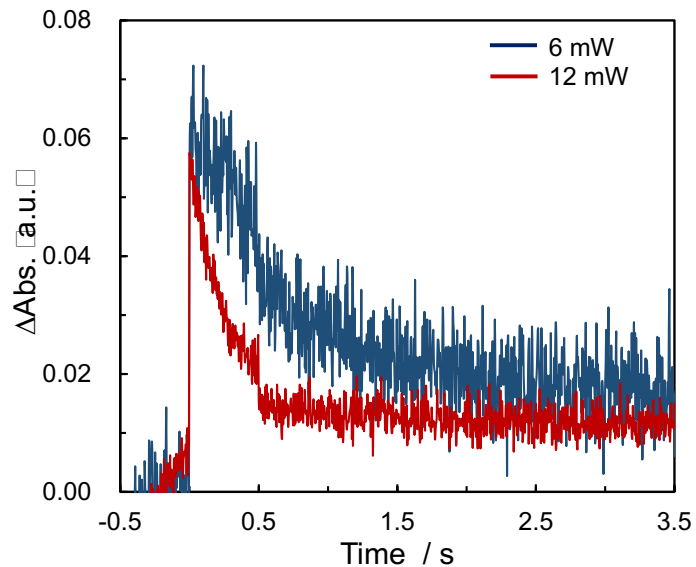


Figure S6. The photochemical bleaching of π -**3a** in toluene under vacuum condition at 293 K.

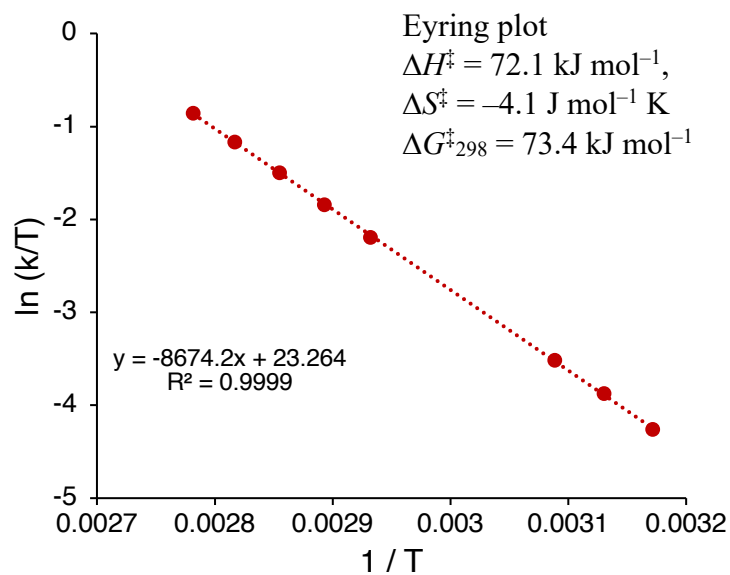


Figure S7. Eyring plots for the fall process of π -**3a** in toluene under argon atmosphere from 315 K to 360 K.

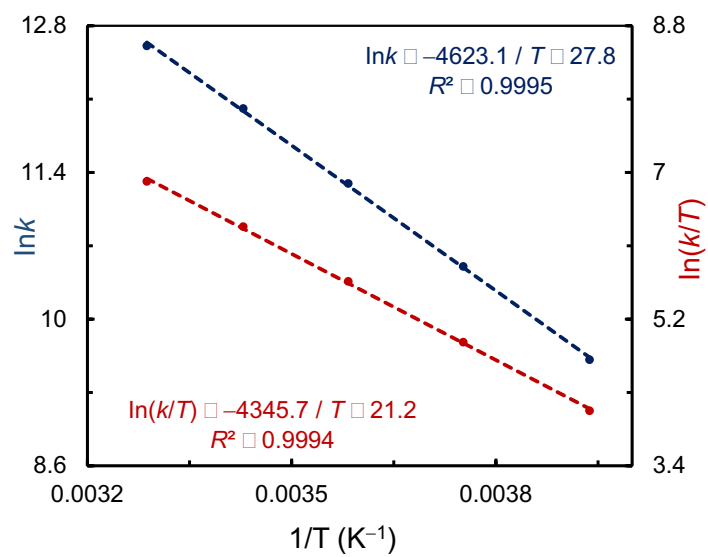


Figure S8. Arrhenius plots (left y axis, blue line) and Eyring plots (right y axis, red line) for the rise process of π -**3a** in toluene under vacuum condition from 243 K to 303 K.

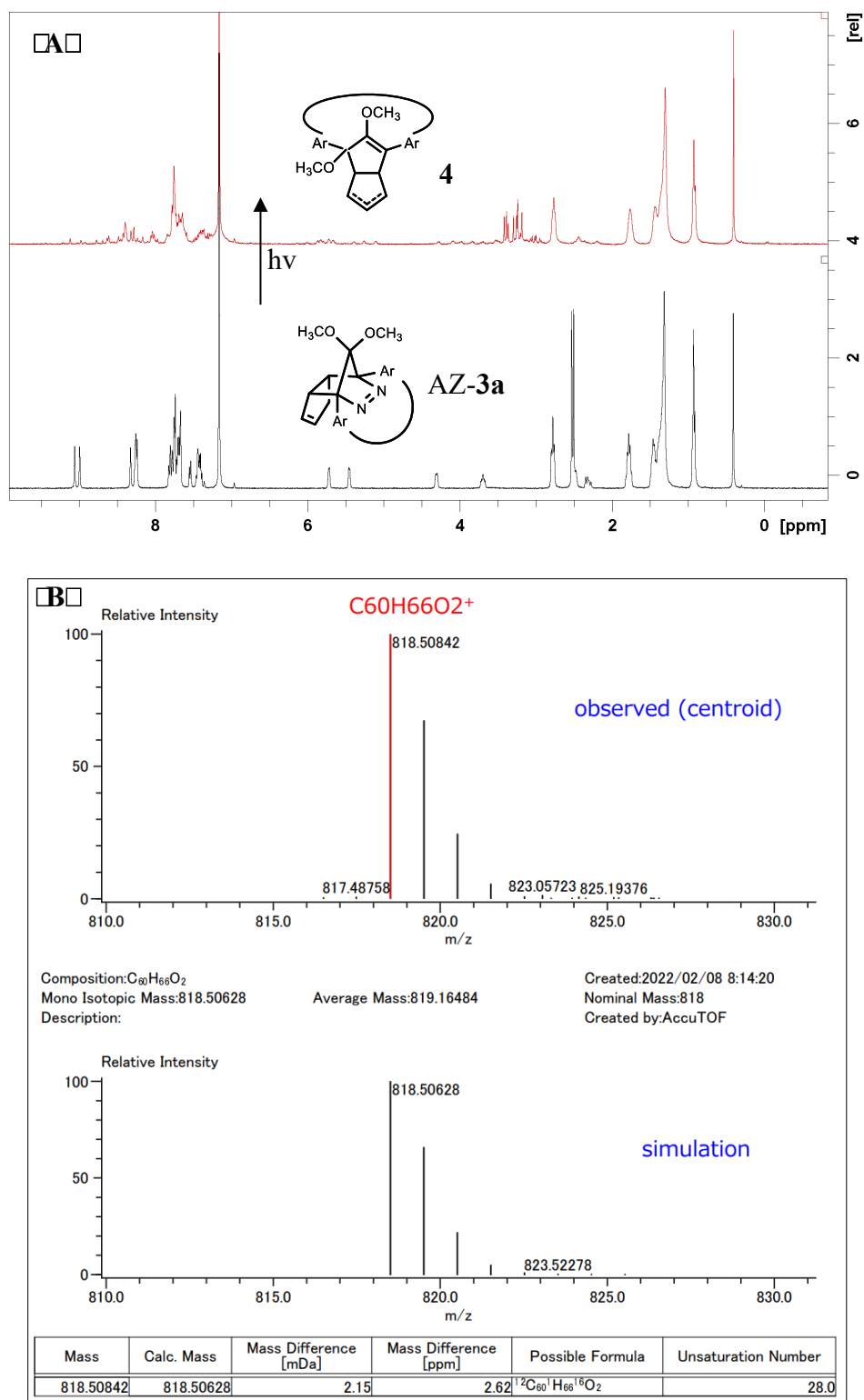


Figure S9. (A) AZ-3a denitrogenation after 10 min irradiation at room temperature in C_6D_6 under vacuum condition. (B) High resolution mass spectrum (FD) of **4**.

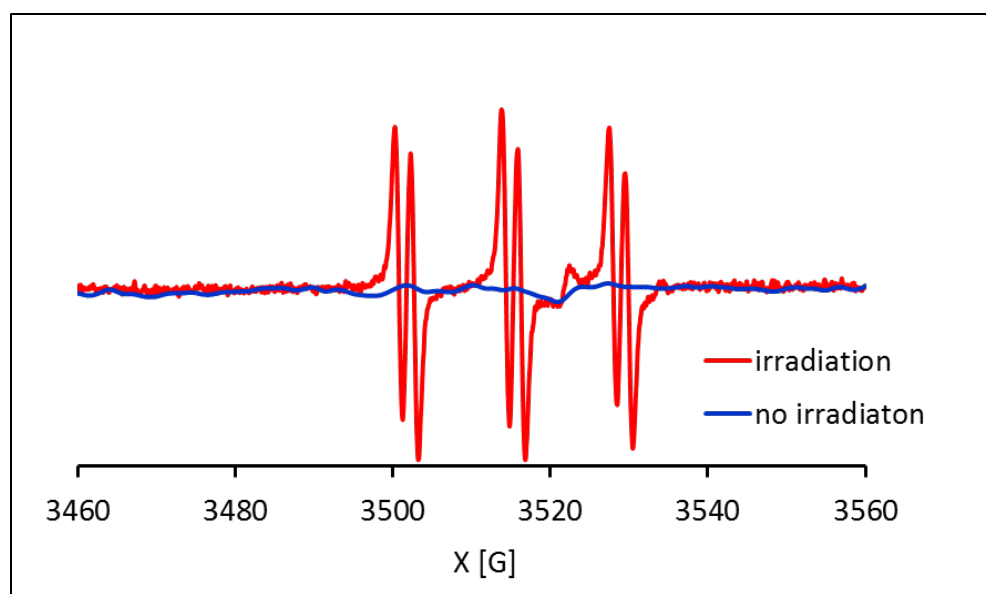


Figure S10. Spin-trapping experiments using electron paramagnetic response (EPR) measurements. EPR spectrum obtained from the photoreaction (Hg lamp) of AZ-3a (4mM) and N-*tert*-buthyl- α -phenylnitron (PBN, 4mM) in benzene solution under vacuum condition at room temperature.

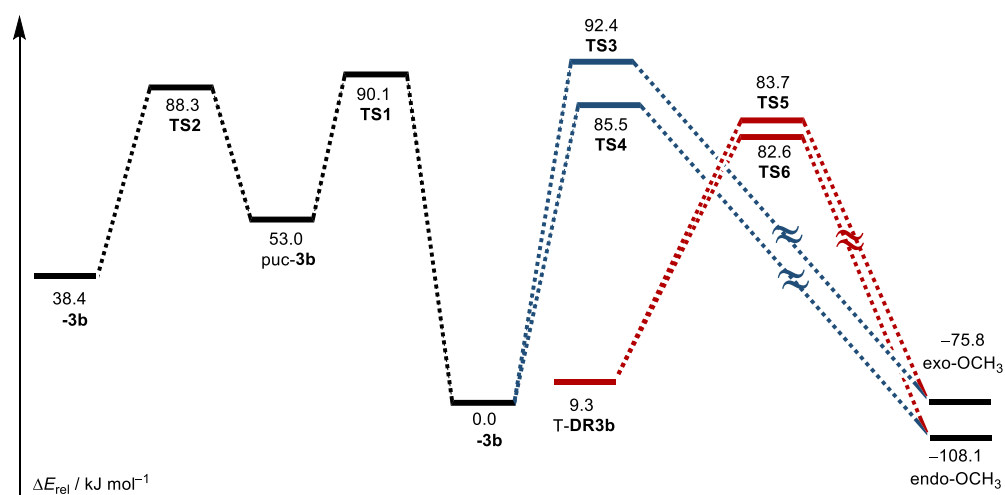


Figure S11. Energy profile of the reaction of the π -3b at [R,U]B3LYP/6-31G(d) level of theory.

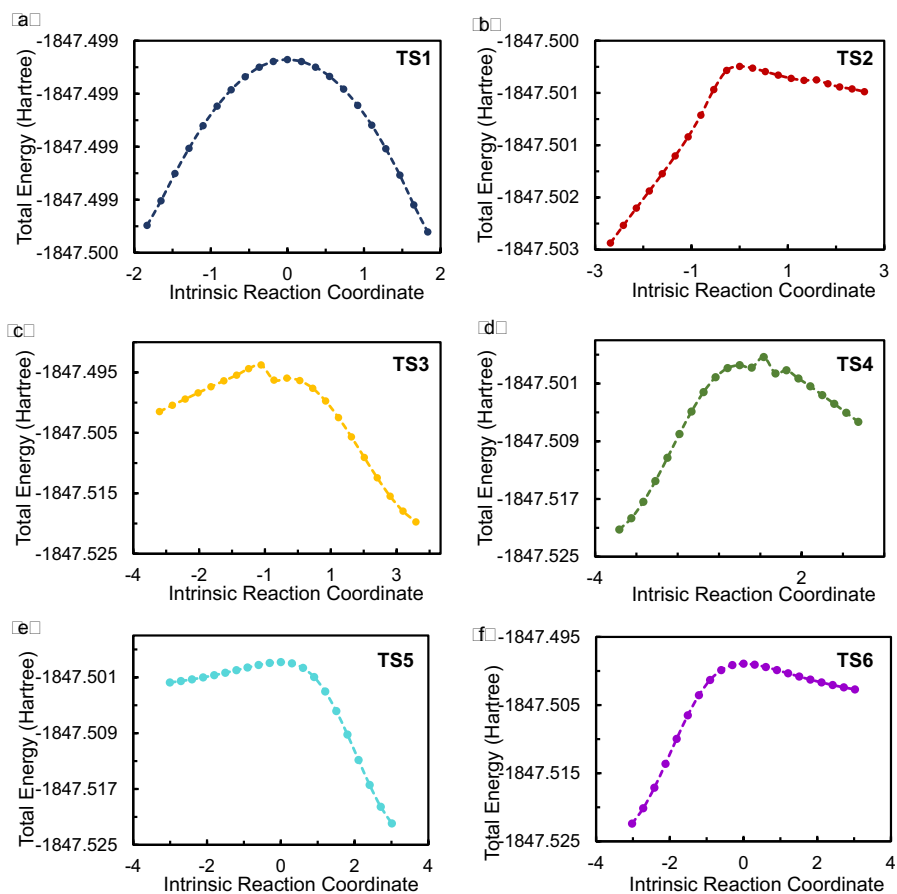
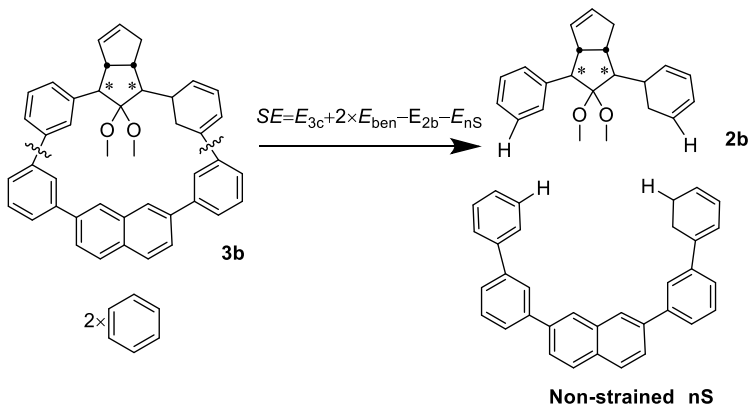


Figure S12. Calculated intrinsic reaction coordinate paths for transition states at $\square R, U \square B3LYP/6-31G \square \square$ level of theory.

Table S2. Molecular strain energy $\square SE \square$ calculated at $\square R, U \square B3LYP/6-31G \square \square$ level of theory.



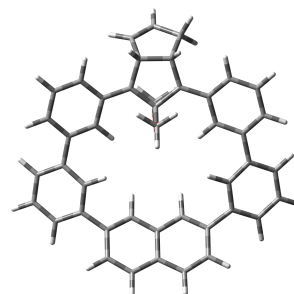
Compounds	SE / kJ mol ⁻¹
π -3b	25.7

Reference

1. Wang, Z.; Akisaka, R.; Yabumoto, S.; Nakagawa, T.; Hatano, S.; Abe, M. Impact of the macrocyclic structure and dynamic solvent effect on the reactivity of a localised singlet diradicaloid with π -single bonding character. *Chem. Sci.*, **2021**, *12*, 613–625.

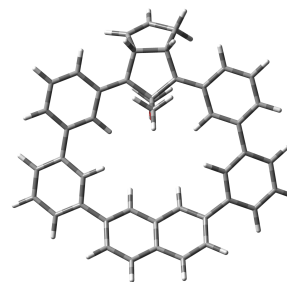
Data S1. Calculated atomic coordinates [in Angstroms] and energies [in Hartrees] for all optimized structures at [R,U]B3LYP/6-31G[d] level of theory.

π -3b [charge = 0, multiplicity=1]			
opt freq ub3lyp/6-31g(d) guess=(mix,always)			
Zero-point correction=	0.652599 (Hartree/Particle)		
Thermal correction to Energy=	0.688505		
Thermal correction to Enthalpy=	0.689449		
Thermal correction to Gibbs Free Energy=	0.586928		
Sum of electronic and zero-point Energies=	-1846.881073		
Sum of electronic and thermal Energies=	-1846.845167		
Sum of electronic and thermal Enthalpies=	-1846.844223		
Sum of electronic and thermal Free Energies=	-1846.946744		
C	4.23233100	-1.72393500	-0.39038800
H	4.55177600	-2.23538000	-1.31293600
C	2.73918600	-1.82943900	-0.21517800
C	2.09494500	-0.46247100	-0.05423800
C	3.25404500	0.50830600	-0.22116600
C	4.56227400	-0.19803200	-0.44644400
H	4.96890600	0.06244200	-1.43363000
C	5.66158500	0.05713600	0.64460300
H	6.61387000	0.36414000	0.18952300
H	5.37683500	0.86276400	1.33339700
C	5.79319000	-1.27147600	1.34467000
H	6.43989500	-1.41058700	2.20775400
C	5.04333300	-2.22703400	0.79351900
H	4.98092900	-3.25528000	1.13766100
O	1.02170400	-0.23098400	-0.97847600
O	1.35848700	-0.29897700	1.16772500
C	1.35777800	-0.30565500	-2.35413800
H	0.43090300	-0.10851500	-2.89795200
H	1.72936700	-1.30053900	-2.63501900
H	2.10315300	0.44975500	-2.63815600
C	2.10494000	-0.42624100	2.36760100
H	2.87253400	0.35469200	2.45744000
H	2.58909600	-1.40777600	2.45080500
H	1.38396500	-0.31088000	3.18034200
C	2.01419500	-3.06393100	-0.28533900
C	0.54460600	-5.47400300	-0.53716700
C	0.62101200	-3.12173300	-0.01417000
C	2.65284200	-4.28121000	-0.65259000
C	1.92271600	-5.45678500	-0.77017200
C	-0.12566700	-4.29686300	-0.15274700
H	0.13928800	-2.21647100	0.32766800
H	3.71729700	-4.29377600	-0.86393400
H	2.42680700	-6.37299900	-1.06700000
H	-0.01248200	-6.39561200	-0.67763700



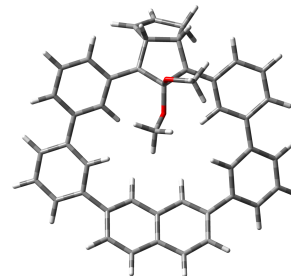
C	3.10273900	1.93342100	-0.29543500
C	2.75789500	4.73387600	-0.56070100
C	1.86343900	2.56423100	-0.00721100
C	4.18052700	2.77432500	-0.68906000
C	4.00095800	4.14613300	-0.81172600
C	1.66724000	3.94196700	-0.15402500
H	1.05478200	1.94212100	0.35030900
H	5.15002100	2.34275200	-0.91615500
H	4.83367200	4.76929700	-1.12784600
H	2.62915800	5.80239400	-0.70641900
C	0.32216200	4.52695200	0.10373900
C	-2.27946000	5.49930900	0.63164700
C	0.12696400	5.84374100	0.55398800
C	-0.81378900	3.73446700	-0.09457200
C	-2.11270900	4.17490200	0.20233900
C	-1.16240400	6.32215300	0.79453700
H	0.98070300	6.48930200	0.74054000
H	-0.68953500	2.74114600	-0.50929200
H	-1.29592300	7.34203000	1.14615800
H	-3.27060600	5.87420900	0.87337200
C	-1.59398600	-4.27164300	0.09586300
C	-4.36621800	-4.07687500	0.61724300
C	-2.29894000	-3.07992800	-0.10573200
C	-2.31920000	-5.38841500	0.54513600
C	-3.69156700	-5.28897200	0.78163400
C	-3.66382300	-2.94123500	0.18937600
H	-1.77362600	-2.22758100	-0.51987600
H	-1.81087100	-6.32968100	0.73476300
H	-4.23686900	-6.16150800	1.13225900
H	-5.42393000	-4.00647900	0.85734200
C	-3.21186600	3.18224000	0.09023600
H	-1.96702800	1.62428400	0.87570900
C	-2.93146400	1.87609100	0.44557400
C	-5.44344700	2.46732500	-0.60855800
C	-3.84318500	0.82486700	0.21575500
C	-4.51428300	3.47016600	-0.41543200
C	-5.12908700	1.10628300	-0.33485800
C	-3.45205400	-0.51136200	0.44060400
H	-4.76274400	4.49206400	-0.69052800
H	-6.97085200	0.18905500	-1.02390400
H	-6.42159800	2.70967300	-1.01812900
C	-4.25019500	-1.58094900	0.08011200
H	-2.47102700	-0.68555500	0.87161900
C	-5.55369200	-1.29873500	-0.42589700
H	-6.20463900	-2.12340400	-0.70495400
C	-5.98099100	0.00063300	-0.61410900

T-DR3b charge 0, multiplicity 3			
opt freq ub3lyp/6-31g(d)			
Zero-point correction=	0.652366 (Hartree/Particle)		
Thermal correction to Energy=	0.688347		
Thermal correction to Enthalpy=	0.689291		
Thermal correction to Gibbs Free Energy=	0.585499		
Sum of electronic and zero-point Energies=	-1846.877523		
Sum of electronic and thermal Energies=	-1846.841542		
Sum of electronic and thermal Enthalpies=	-1846.840598		
Sum of electronic and thermal Free Energies=	-1846.944390		
C	2.10014600	-0.48337800	-0.04956200
C	3.26566400	0.49007800	-0.21808400
C	2.72284300	-1.86701300	-0.21367100
C	4.21668200	-1.76474100	-0.39992400
H	4.53093500	-2.29097200	-1.31527100
C	4.56071000	-0.23614400	-0.47121400
H	4.94857900	0.02254800	-1.46641300
C	1.98569400	-3.08812400	-0.27645500
C	0.48906100	-5.48515900	-0.51903600
C	0.58865400	-3.12972100	-0.01078400
C	2.61207800	-4.31766500	-0.63223400
C	1.86907900	-5.48435300	-0.74606500
C	-0.16935600	-4.29787200	-0.14436500
H	0.11467500	-2.21749900	0.32317600
H	3.67767600	-4.34395600	-0.83673500
H	2.36402800	-6.40825200	-1.03456400
H	-0.07758200	-6.40122100	-0.65685900
C	3.12172800	1.91071800	-0.28818300
C	2.79349200	4.71590300	-0.54876400
C	4.20580700	2.74849600	-0.67894800
C	1.88434600	2.55022300	-0.00020800
C	1.69779000	3.92929000	-0.14432500
C	4.03363700	4.12050900	-0.79999600
H	5.17335500	2.31202400	-0.90487200
H	1.07042900	1.93317800	0.35407200
H	4.87022600	4.73944400	-1.11437500
H	2.67110400	5.78524900	-0.69333500
C	-1.63812200	-4.25746300	0.10050000
C	-4.41016000	-4.03681300	0.61514400
C	-2.33228700	-3.06024600	-0.10582500
C	-2.37454000	-5.36628800	0.55148500
C	-3.74652300	-5.25427200	0.78443200
C	-3.69653200	-2.90872200	0.18603000
H	-1.79873100	-2.21360800	-0.52109000
H	-1.87499500	-6.31138400	0.74515800
H	-4.30019400	-6.12106200	1.13621100



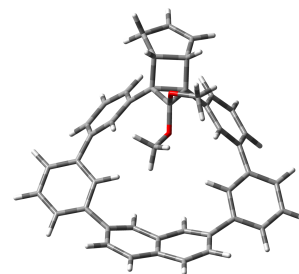
H	-5.46773300	-3.95626400	0.85266600
C	0.35669800	4.52387700	0.11279300
C	-2.23787900	5.51798400	0.63631900
C	0.17161000	5.84135700	0.56543700
C	-0.78565200	3.74169000	-0.08945200
C	-2.08143900	4.19303300	0.20490200
C	-1.11406700	6.33058900	0.80387300
H	1.03052100	6.47903100	0.75523200
H	-0.66899300	2.74813800	-0.50582500
H	-1.23948100	7.35088800	1.15726300
H	-3.22640300	5.90100100	0.87606000
O	1.03909200	-0.24273100	-0.97895300
O	1.37352200	-0.31617200	1.17113200
C	5.68603900	0.01012800	0.59554500
H	6.63752800	0.28713900	0.11922700
H	5.43177500	0.83362100	1.27464600
C	5.02799400	-2.26034100	0.78781800
H	4.95292400	-3.28244400	1.14739200
C	5.80004200	-1.30931100	1.31499500
H	6.45943200	-1.44837400	2.16830800
C	-4.27026600	-1.54334400	0.07345600
H	-2.48654600	-0.66444500	0.87239700
C	-3.46396300	-0.48098700	0.43702900
C	-5.98372400	0.05381900	-0.62795100
C	-3.84192300	0.85873000	0.21059200
C	-5.56905300	-1.24938000	-0.43790900
C	-5.12299900	1.15173800	-0.34522100
C	-2.92184800	1.90177300	0.44425700
H	-6.22630900	-2.06817800	-0.71944000
H	-6.39831800	2.76670700	-1.03346000
H	-6.97016500	0.25121700	-1.04174800
C	-3.18906700	3.21041200	0.08809700
H	-1.96143200	1.64170300	0.87848000
C	-4.48677900	3.51007800	-0.42281100
H	-4.72492000	4.53423200	-0.69863000
C	-5.42404000	2.51558600	-0.61996300
C	2.12210200	-0.44805500	2.37065700
H	2.89383500	0.32856100	2.45687800
H	2.59868700	-1.43314000	2.45135200
H	1.40277500	-0.32818300	3.18394600
C	1.37699400	-0.32113800	-2.35537400
H	0.45228600	-0.11634800	-2.89963500
H	1.73970600	-1.31974000	-2.63308400
H	2.12838500	0.42873400	-2.63678800

puc-3b charge 0, multiplicity 1			
opt freq ub3lyp/6-31g(d) guess=(mix,always)			
Zero-point correction=		0.652587 (Hartree/Particle)	
Thermal correction to Energy=		0.688434	
Thermal correction to Enthalpy=		0.689378	
Thermal correction to Gibbs Free Energy=		0.586870	
Sum of electronic and zero-point Energies=		-1846.860889	
Sum of electronic and thermal Energies=		-1846.825043	
Sum of electronic and thermal Enthalpies=		-1846.824099	
Sum of electronic and thermal Free Energies=		-1846.926607	
C	4.79752100	2.04490100	0.39121200
C	3.62549500	1.54639300	-0.25223600
C	2.53924000	3.73066100	-0.08879600
C	5.83497600	1.11391700	0.68083400
H	6.74419400	1.47155400	1.15890600
C	4.17633500	-2.17837600	-0.36837200
C	-4.46658600	-1.13120700	-0.09412200
H	-5.23287900	-1.00407100	0.68624100
C	0.07407900	3.80677200	-0.04529000
H	0.13411800	2.81578600	0.38612000
C	-1.77961300	-4.74406100	0.37293400
H	-2.81893100	-4.97947000	0.57850800
C	-4.70375000	1.85910700	-0.07432800
H	-5.59955300	1.25054900	-0.00787800
C	-3.69567000	4.06073200	-0.20874500
H	-3.82397500	5.13809500	-0.23588800
C	-2.21823100	-0.94761500	0.86228600
C	0.51056200	-5.49710200	0.12545200
H	1.23020800	-6.30975200	0.14598500
C	4.61494300	-4.43409300	-1.12057900
C	0.94212500	-4.17759200	-0.09861000
C	-5.14374200	-1.03693800	-1.50228000
H	-6.22710200	-0.87163100	-1.41861300
H	-4.75029700	-0.19750200	-2.08908800
C	-2.37409200	-2.33967400	0.28345300
C	-4.14756500	-3.18178400	-1.35102300
H	-3.80825300	-4.17710500	-1.62043200
C	-2.29782200	2.10026300	-0.13510700
H	-1.32970300	1.62728700	-0.18601400
C	-2.97552200	0.04661700	2.97414000
H	-2.33810300	0.92389600	2.81124000
H	-2.95229400	-0.22399200	4.03330400
H	-4.00313600	0.31173700	2.69569300
C	3.52052300	0.17176500	-0.55096300
H	2.62313900	-0.17320200	-1.05601600
C	-4.82440800	3.24089700	-0.14885900



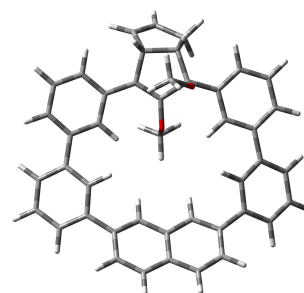
H	-5.81343400	3.69211800	-0.14835900
C	2.37562200	-3.86220400	-0.35974100
C	-0.00707100	-0.47672100	1.77184500
H	-0.28458000	0.13456500	2.63632200
H	0.94122200	-0.10742500	1.37261800
H	0.11047100	-1.51402700	2.09644600
C	2.86340300	-2.58119200	-0.08120000
H	2.20638300	-1.86264200	0.39083200
C	4.49266200	-0.73865400	-0.18001400
C	-0.84088500	-5.76828500	0.34506100
H	-1.16133700	-6.79291600	0.51570300
C	4.82142500	3.42654100	0.73410400
H	5.70768500	3.83373200	1.21562800
C	-4.83650000	-2.36381400	-2.14626700
H	-5.14716000	-2.60063600	-3.16102100
C	5.68544600	-0.23345200	0.41620700
H	6.47084900	-0.92821100	0.70266400
C	-1.18209600	4.33942600	-0.35875800
C	-3.42271400	1.24413000	-0.02573100
C	5.06743300	-3.13593300	-0.87102500
H	6.09227900	-2.86042000	-1.10574700
C	-3.31515200	-0.18660700	0.14479900
C	1.27722200	4.49508500	-0.26409400
C	-1.38511000	-3.39170200	0.19845100
C	3.72737100	4.24001300	0.51384600
H	3.75537400	5.27731200	0.83763900
C	-2.40683700	3.49600600	-0.22320200
C	-0.01615500	-3.15200000	-0.07068000
H	0.27522600	-2.13806500	-0.30807000
C	3.28528000	-4.79324200	-0.89199300
H	2.95395700	-5.79455000	-1.15101000
C	1.21113800	5.82113000	-0.71274200
H	2.12292600	6.38420000	-0.89378600
C	-1.21265300	5.66237400	-0.83616500
H	-2.15424600	6.12197600	-1.12013600
C	-0.03323300	6.39423300	-0.98310700
C	-3.84141900	-2.56350000	-0.00100700
H	-4.30447500	-3.14913300	0.81203900
C	2.54111200	2.41234600	-0.50490300
H	1.67330500	2.00589100	-1.01545100
O	-2.51357800	-1.10394600	2.26942500
O	-0.94288400	-0.34294000	0.69791200
H	-0.08380600	7.41526300	-1.35246900
H	5.30109200	-5.16949900	-1.53272800

σ-3b charge = 0, multiplicity = 1			
opt freq rb3lyp/6-31g(d)			
Zero-point correction=	0.654906 (Hartree/Particle)		
Thermal correction to Energy=	0.690266		
Thermal correction to Enthalpy=	0.691210		
Thermal correction to Gibbs Free Energy=	0.590490		
Sum of electronic and zero-point Energies=	-1846.866459		
Sum of electronic and thermal Energies=	-1846.831099		
Sum of electronic and thermal Enthalpies=	-1846.830154		
Sum of electronic and thermal Free Energies=	-1846.930875		
C	-4.31164400	-0.92579100	1.44303100
H	-4.23969800	-1.29375600	2.46794400
C	-2.92082100	-0.88329700	0.75475600
C	-2.05528400	0.01784800	1.61224800
C	-3.00348900	0.72080600	0.66802600
C	-4.42886800	0.63048900	1.28593400
H	-4.56755000	1.22666700	2.19707200
C	-2.41613400	-1.87889600	-0.23953300
C	-1.50424900	-3.79228500	-2.11347100
C	-1.26920500	-2.64635700	-0.00531300
C	-3.09647300	-2.08425700	-1.45409100
C	-2.66330200	-3.05342500	-2.35906300
C	-0.76856600	-3.56559200	-0.94327300
H	-0.72773900	-2.52118200	0.92197100
H	-3.95967000	-1.47153900	-1.69593200
H	-1.13564200	-4.49414100	-2.85707400
C	-2.56867300	1.73752300	-0.34300400
C	-1.75323200	3.73380700	-2.17499200
C	-1.42499500	2.51820100	-0.11989300
C	-3.30084800	1.98097800	-1.51912300
C	-2.91050700	2.98702100	-2.40366700
C	-0.97412000	3.48051200	-1.03811600
H	-0.84056600	2.34403800	0.77293700
H	-4.17776100	1.38302900	-1.74199000
H	-3.49613100	3.16616300	-3.30193400
H	-1.42453200	4.46870200	-2.90541800
C	3.32265500	-4.84342100	-0.46621100
C	1.59896300	-3.17340700	-0.44622200
C	1.00058300	-5.44508300	-0.92229700
C	2.34442700	-5.79929200	-0.76065000
C	2.95897300	-3.49660500	-0.32501300
H	1.29591100	-2.14145800	-0.31892600
H	0.26179300	-6.20763400	-1.15523300
H	4.36460800	-5.14203100	-0.38303000
C	0.38334300	4.06982800	-0.85906000
C	3.06299300	4.93250400	-0.58791200



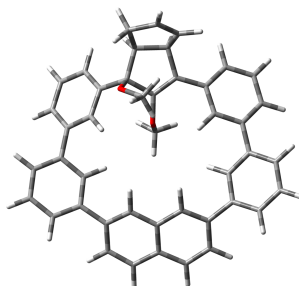
C	0.72093700	5.41539700	-1.06958000
C	1.41185900	3.18807200	-0.51003000
C	2.75579700	3.57736200	-0.39987600
C	2.04632400	5.83482200	-0.91685400
H	-0.04938600	6.13651000	-1.33091900
H	1.15499100	2.14967700	-0.33619000
H	2.29387300	6.88210500	-1.07078900
H	4.09158500	5.27670200	-0.51484900
C	3.87334900	-2.34797100	-0.07510300
H	2.84524500	-1.15589700	-1.52578100
C	3.58333500	-1.16421100	-0.72947700
C	5.48557900	-1.13217200	1.30408100
C	4.09850900	0.07567900	-0.29359000
C	4.89667400	-2.32152600	0.91864300
C	5.05200000	0.11394900	0.76688300
C	3.50651400	1.27326000	-0.75009500
H	6.15777300	1.45496200	2.06793800
H	6.23570700	-1.13380700	2.09181100
C	3.72704900	2.48406500	-0.11976300
H	2.76724600	1.20615900	-1.54254400
C	4.75072900	2.53754600	0.87255300
C	5.40941100	1.39346100	1.28092700
H	-3.21104800	-3.20474600	-3.28570100
H	2.63736700	-6.83939600	-0.87922700
C	-5.59106600	0.79735100	0.33923400
H	-5.91762200	1.77131200	-0.01375000
C	-6.17321600	-0.36485900	0.02664000
H	-7.04418400	-0.46758900	-0.61638200
C	-5.51762800	-1.54714200	0.70195300
H	-6.20828800	-2.02352700	1.41307600
H	-5.22516500	-2.33427700	-0.00297200
H	4.97151100	3.48553700	1.35676300
H	5.17228300	-3.24474500	1.42221400
O	-2.27161100	0.03440200	2.99266700
O	-0.69914900	0.12699200	1.26820200
C	0.22386000	-0.49614700	2.16339200
H	0.40227700	0.12129600	3.05082200
H	1.15847200	-0.59574900	1.60778800
H	-0.11244000	-1.48238200	2.49839500
C	-2.09337300	1.31452600	3.59801700
H	-2.40399400	1.20748800	4.64000400
H	-2.71128300	2.07692000	3.10853100
H	-1.04633200	1.63971600	3.56607000
C	0.60757700	-4.10898700	-0.75631800

exo-OCH ₃ □ charge □ 0, multiplicity □ 1 □			
opt freq rb3lyp/6-31g(d)			
Zero-point correction=	0.655456 (Hartree/Particle)		
Thermal correction to Energy=	0.691100		
Thermal correction to Enthalpy=	0.692044		
Thermal correction to Gibbs Free Energy=	0.589988		
Sum of electronic and zero-point Energies=	-1846.909954		
Sum of electronic and thermal Energies=	-1846.874311		
Sum of electronic and thermal Enthalpies=	-1846.873366		
Sum of electronic and thermal Free Energies=	-1846.975422		
C	-2.11135500	-0.88135200	0.00723500
C	-3.17842100	0.00664200	0.66366800
C	-2.41498300	-2.20306400	0.06104800
C	-3.89685000	-2.36145900	0.39737100
H	-4.02676000	-2.91527600	1.33716000
C	-4.43428000	-0.88920800	0.48383800
H	-5.10019800	-0.76946300	1.34362300
C	-1.50299000	-3.36488300	-0.03709400
C	0.26923300	-5.57311300	0.05411900
C	-0.11301900	-3.20965800	-0.20029200
C	-1.98292900	-4.67560400	0.15663300
C	-1.10499200	-5.75845600	0.19112400
C	0.78940200	-4.28255600	-0.13778700
H	0.26945600	-2.21221300	-0.35052900
H	-3.04130000	-4.85555900	0.30688200
H	-1.49837500	-6.75904600	0.35073000
H	0.93684800	-6.42661900	0.12780300
C	-3.30898000	1.48200300	0.25140300
C	-3.45298200	4.27929300	-0.08162100
C	-4.54269200	2.14529300	0.30094200
C	-2.16147400	2.25493500	0.04382400
C	-2.19810200	3.65185300	-0.11854400
C	-4.60865400	3.52768800	0.12142700
H	-5.45530500	1.59732000	0.50731900
H	-1.21176800	1.74333600	0.03523000
H	-5.57245600	4.02810600	0.16837600
H	-3.53006300	5.35853600	-0.17507000
C	2.25222900	-4.02217900	-0.26171100
C	4.99162500	-3.37823000	-0.57081400
C	2.75945100	-2.76330500	0.08228300
C	3.16571700	-4.97722200	-0.74066600
C	4.51922600	-4.65904900	-0.86756500
C	4.10037500	-2.39762400	-0.11260600
H	2.09394000	-2.03325100	0.52655500
H	2.81956300	-5.96244500	-1.03991200
H	5.21011200	-5.41046700	-1.24112100



H	6.03736900	-3.13233900	-0.73570400
C	-0.92917500	4.41349300	-0.30960400
C	1.54903800	5.72469400	-0.74231800
C	-0.88014600	5.69681800	-0.88320000
C	0.29187400	3.83549000	0.05988300
C	1.53487900	4.43423000	-0.19560900
C	0.34148200	6.34674600	-1.06856400
H	-1.79314000	6.18626100	-1.20915500
H	0.29203200	2.88080900	0.57025400
H	0.35368100	7.33911100	-1.51204900
H	2.49307500	6.21972500	-0.95458000
O	-2.83316700	0.19764600	2.07137100
O	-0.89275900	-0.36757000	-0.38153000
C	-5.23834400	-0.67760100	-0.83674900
H	-6.19446100	-0.17040800	-0.65923900
H	-4.69237300	-0.04701500	-1.55243400
C	-4.70982700	-2.97748500	-0.72709300
H	-4.66928500	-4.03323900	-0.97611600
C	-5.42690600	-2.06722500	-1.38504400
H	-6.05822900	-2.27249000	-2.24571800
C	4.45833300	-0.97033100	0.09730300
H	2.60604400	-0.33124400	-0.76386400
C	3.52440200	-0.02173800	-0.27498200
C	5.89203800	0.83598000	0.91260500
C	3.70023700	1.35292800	-0.01546900
C	5.67584500	-0.50820800	0.67995500
C	4.90346500	1.81145800	0.59835800
C	2.66653000	2.26797500	-0.30170300
H	6.43083500	-1.23197000	0.97618000
H	5.91904400	3.58057400	1.33788500
H	6.82153800	1.15944500	1.37578500
C	2.74685800	3.60535600	0.03833500
H	1.77030200	1.88824200	-0.78230300
C	3.96567900	4.07088800	0.61460900
H	4.05892800	5.12007400	0.88319000
C	5.00845200	3.20376400	0.87747700
C	-0.76059300	-0.24336300	-1.80608900
H	-1.51106900	0.44750000	-2.20682000
H	-0.85878400	-1.21902500	-2.29533300
H	0.23802500	0.15991200	-1.98979400
C	-2.70507500	-0.91582700	2.93781000
H	-2.36446500	-0.50204400	3.89111900
H	-1.96521700	-1.64724600	2.58876200
H	-3.66173600	-1.43092600	3.10739600

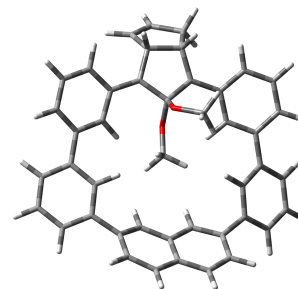
endo-OCH ₃ □ charge □ 0, multiplicity □ 1 □			
opt freq rb3lyp/6-31g(d)			
Zero-point correction=		0.655377 (Hartree/Particle)	
Thermal correction to Energy=		0.691001	
Thermal correction to Enthalpy=		0.691945	
Thermal correction to Gibbs Free Energy=		0.589898	
Sum of electronic and zero-point Energies=		-1846.922258	
Sum of electronic and thermal Energies=		-1846.886633	
Sum of electronic and thermal Enthalpies=		-1846.885689	
Sum of electronic and thermal Free Energies=		-1846.987736	



C	-2.28177000	-0.02620400	-0.15378100
C	-3.03084500	-1.25938700	0.35939800
C	-3.05898700	1.05944400	-0.36480700
C	-4.52219600	0.67332000	-0.23458600
H	-5.04329900	0.88320300	-1.18284300
C	-4.52186900	-0.86148600	0.07482500
H	-4.86139700	-1.39690100	-0.81131000
C	-2.61767000	2.43951500	-0.65738300
C	-1.73791700	5.07599400	-1.16657000
C	-1.34288300	2.88033600	-0.26259500
C	-3.46137700	3.36955700	-1.29248800
C	-3.02027900	4.67064000	-1.53744800
C	-0.87333800	4.17642600	-0.52185100
H	-0.70956000	2.18845900	0.27658400
H	-4.45981200	3.07710400	-1.60385000
H	-3.68053800	5.37355200	-2.03882900
H	-1.39995300	6.08175000	-1.40068500
C	-2.54283700	-2.57221900	-0.28537900
C	-1.43748600	-4.93973200	-1.36789000
C	-3.27234100	-3.35667600	-1.18677300
C	-1.25822300	-3.01250600	0.06529100
C	-0.67274800	-4.16174100	-0.48486500
C	-2.72763000	-4.53720900	-1.70432600
H	-4.27541400	-3.07792600	-1.49127100
H	-0.70342800	-2.44179600	0.79749700
H	-3.31649100	-5.13723200	-2.39335800
H	-1.01079000	-5.83444100	-1.81382400
C	0.51812400	4.53255100	-0.12987600
C	3.18485300	5.03814500	0.66159700
C	1.49758500	3.53297000	-0.15190300
C	0.91071500	5.81819800	0.27684800
C	2.23377900	6.06185200	0.65299000
C	2.81845000	3.74187500	0.27066200
H	1.23286300	2.55617400	-0.53756600
H	0.18105300	6.62208000	0.32338300
H	2.52150200	7.06019400	0.97235200

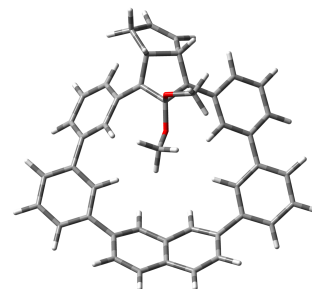
H	4.19764600	5.24118200	0.99970100
C	0.74999700	-4.46328500	-0.16857800
C	3.48219000	-4.83809800	0.43542700
C	1.25541300	-5.75181800	0.06406400
C	1.64679000	-3.39097100	-0.11464800
C	3.00482500	-3.53900800	0.20337900
C	2.60995500	-5.92716300	0.35842800
H	0.59008700	-6.61079100	0.03697300
H	1.27506200	-2.40642000	-0.36931700
H	2.98888500	-6.92770200	0.55060700
H	4.52492200	-4.99924200	0.69613800
O	-0.90873800	-0.07633700	-0.22100100
O	-2.75922300	-1.47943200	1.76333900
C	-5.51812300	-1.06786200	1.24577100
H	-6.42596500	-1.58984500	0.91177700
H	-5.07869100	-1.67605900	2.04371100
C	-5.30130800	1.27603100	0.91724100
H	-5.40751900	2.34827600	1.05516900
C	-5.83991300	0.33749100	1.70037800
H	-6.45536000	0.53809200	2.57427200
C	3.72301700	2.56246900	0.28426400
H	2.16697100	1.27293800	0.99821500
C	3.18711000	1.33465600	0.63054000
C	5.82422600	1.44262000	-0.26684600
C	3.89846300	0.12736600	0.44995500
C	5.08674400	2.60124000	-0.13172100
C	5.24327100	0.16589800	-0.02761000
C	3.24969300	-1.11627800	0.61866700
H	5.53486800	3.55674900	-0.39167900
H	6.91311800	-1.07319100	-0.64515500
H	6.85353600	1.49514500	-0.61448200
C	3.84097400	-2.31053800	0.24513500
H	2.23241100	-1.10757500	0.99817000
C	5.20214400	-2.27080400	-0.18164500
H	5.69527600	-3.19628900	-0.46679300
C	5.88371900	-1.07641600	-0.29383800
C	-2.46567700	-0.36868400	2.60456000
H	-3.26733300	0.37734100	2.62051300
H	-1.52434900	0.12066700	2.32309900
H	-2.35445400	-0.79375300	3.60624100
C	-0.37015800	-0.05618700	-1.55244300
H	0.71480600	-0.12034400	-1.44319600
H	-0.63142700	0.87274100	-2.07038300
H	-0.73231500	-0.91589300	-2.12600400

TS1 charge 0, multiplicity1, imaginary frequencies -75.63			
opt=(TS,calcfc,noeig,MaxCycle=300) guess=(mix,always) freq			
ub3lyp/6-31g(d) iop(1/8=3) scfcyc=300 test			
Zero-point correction=		0.652398 (Hartree/Particle)	
Thermal correction to Energy=		0.687291	
Thermal correction to Enthalpy=		0.688235	
Thermal correction to Gibbs Free Energy=		0.588759	
Sum of electronic and zero-point Energies=		-1846.846737	
Sum of electronic and thermal Energies=		-1846.811845	
Sum of electronic and thermal Enthalpies=		-1846.810900	
Sum of electronic and thermal Free Energies=		-1846.910376	
C	2.42989200	4.45014100	0.57284300
C	1.90460200	3.45252700	-0.30363600
C	-0.32816500	4.45918500	-0.16942000
C	3.77031600	4.29495200	1.02765100
H	4.19462400	5.05634300	1.67832700
C	4.61086900	0.80346500	-0.29575200
C	-2.84569900	-3.66829800	0.38254700
H	-3.29691100	-3.87250400	1.36522600
C	-2.32370100	2.99757300	-0.09709800
H	-1.67486900	2.28035600	0.38766700
C	1.52425900	-4.90827400	0.41870000
H	0.84918300	-5.72913500	0.63960500
C	-4.86707800	-1.40831400	0.55224900
H	-5.17744500	-2.41148600	0.82681800
C	-5.42803100	0.94834900	0.32613600
H	-6.15801200	1.74539100	0.43448200
C	-1.02151900	-1.98183200	0.35515100
C	3.80814200	-4.08068800	0.27877300
H	4.87129300	-4.26391200	0.40403400
C	6.41985700	-0.68377800	-0.90467100
C	3.34334100	-2.79980200	-0.07464900
C	-3.76797100	-4.25169400	-0.74266800
H	-4.62088400	-4.80083200	-0.31884200
H	-4.19260300	-3.46108300	-1.37350600
C	-0.41213600	-3.35445400	0.15199100
C	-1.63988600	-5.26512400	-0.98983600
H	-0.82325600	-5.85780000	-1.39131000
C	-3.20237700	0.18147300	-0.17009900
H	-2.22382100	0.37328700	-0.57303100
C	-1.35326800	-0.71856500	2.46766400
H	-1.51548700	0.26866300	2.01792600
H	-0.78998400	-0.59982600	3.39708000
H	-2.33093100	-1.15991700	2.69338200
C	2.71573000	2.34553400	-0.65543900
H	2.31199700	1.60584400	-1.33688000



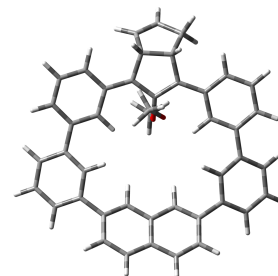
C	-5.79257100	-0.37154900	0.61159500
H	-6.81394500	-0.58823300	0.91431200
C	4.26704000	-1.64584700	-0.30189100
C	0.18916100	0.06576300	-0.32034600
H	-0.42424700	0.95805900	-0.18229500
H	0.85587600	0.23939900	-1.16653600
H	0.76885200	-0.08968000	0.58993600
C	3.81759900	-0.33481100	-0.07771300
H	2.82107400	-0.18387000	0.31224200
C	3.96621000	2.13668500	-0.10101500
C	2.89727400	-5.11892500	0.49722100
H	3.26895000	-6.10561600	0.76209300
C	1.55708600	5.49550600	0.98825000
H	1.94624700	6.27740400	1.63657500
C	-2.85823800	-5.16706200	-1.52113800
H	-3.18527500	-5.67796100	-2.42347800
C	4.50750700	3.16868700	0.72081100
H	5.49907800	3.04080500	1.14741600
C	-3.65363600	2.63540200	-0.35986200
C	-3.51455700	-1.15825100	0.18876000
C	5.94089700	0.61213600	-0.69503800
H	6.58740400	1.46793600	-0.87149500
C	-2.51908500	-2.20357700	0.24773000
C	-1.79295500	4.26377000	-0.39070700
C	1.00407800	-3.62261300	0.10885000
C	0.21971100	5.49279600	0.64570300
H	-0.43976900	6.26177200	1.03982700
C	-4.11060800	1.24221700	-0.07124700
C	1.96107800	-2.61727100	-0.18856000
H	1.60181600	-1.67381200	-0.55252800
C	5.59770000	-1.79905200	-0.72729300
H	5.99127200	-2.78933900	-0.93876300
C	-2.66410400	5.23826200	-0.89660300
H	-2.29318500	6.23201600	-1.13381300
C	-4.49398300	3.62958400	-0.88830800
H	-5.52770700	3.39665000	-1.12880900
C	-4.00301700	4.91417800	-1.13427300
C	-1.47526000	-4.41752200	0.25987400
H	-1.29136300	-5.06229800	1.13583200
C	0.54074500	3.51346200	-0.68584500
H	0.15697000	2.75059800	-1.35518600
O	-0.55986500	-1.57313200	1.65263800
O	-0.59918200	-1.05868400	-0.66413500
H	-4.66944600	5.66813500	-1.54527400
H	7.44538500	-0.82690600	-1.23540800

TS2 □charge □ 0, multiplicity□1, imaginary frequencies□ −154.73□			
opt=(calcfc,ts,noeigen) freq ub3lyp/6-31g(d) guess=(mix,always)			
Zero-point correction=	0.652937 (Hartree/Particle)		
Thermal correction to Energy=	0.687965		
Thermal correction to Enthalpy=	0.688909		
Thermal correction to Gibbs Free Energy=	0.589141		
Sum of electronic and zero-point Energies=	-1846.847458		
Sum of electronic and thermal Energies=	-1846.812430		
Sum of electronic and thermal Enthalpies=	-1846.811486		
Sum of electronic and thermal Free Energies=	-1846.911255		
C	-4.48311500	-0.94464000	1.05075200
H	-4.53997600	-1.01567300	2.14234200
C	-3.03623300	-1.20194100	0.65699300
C	-2.28140700	-0.03787900	1.25053200
C	-3.11470700	0.94024300	0.45921900
C	-4.58001300	0.53723800	0.56479000
H	-5.13648000	1.21402100	1.24184400
C	-2.49489400	-2.30136600	-0.13221000
C	-1.47374000	-4.48324100	-1.64087300
C	-1.22108500	-2.85705300	0.12615200
C	-3.22536400	-2.86661200	-1.20799400
C	-2.73159000	-3.95068600	-1.92516200
C	-0.68867100	-3.91339200	-0.62765800
H	-0.63482500	-2.47295800	0.94760200
H	-4.15643700	-2.41225200	-1.51906600
H	-1.07392900	-5.29484500	-2.24256400
C	-2.68761300	2.19209900	-0.14348500
C	-1.94092300	4.71851400	-1.22128300
C	-1.34142300	2.63968100	-0.15810800
C	-3.65321600	3.09268700	-0.66823900
C	-3.27826000	4.32536300	-1.19061000
C	-0.94935000	3.87668600	-0.68653300
H	-0.60047600	1.98648600	0.27222300
H	-4.70569800	2.84276900	-0.63977400
H	-4.04168600	4.98863700	-1.58945000
H	-1.67056400	5.66779000	-1.67338800
C	3.49122800	-4.87982100	-0.23116000
C	1.67962600	-3.30300300	-0.29840100
C	1.19127700	-5.64552100	-0.47197800
C	2.55792100	-5.91287400	-0.35555500
C	3.05908000	-3.54552400	-0.21500800
H	1.32467700	-2.28007900	-0.26969000
H	0.48381300	-6.46533300	-0.56541500
H	4.55172200	-5.11185600	-0.17680100
C	0.48777200	4.28075700	-0.67155100



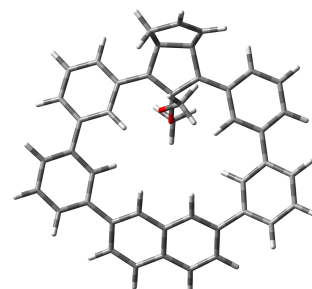
C	3.23571700	4.95987200	-0.57004200
C	0.90047500	5.62474500	-0.70300500
C	1.48821500	3.30567100	-0.59089400
C	2.85736700	3.61003500	-0.52950100
C	2.25686900	5.95104800	-0.66119700
H	0.16405000	6.42239700	-0.72797600
H	1.19431700	2.26514200	-0.54780800
H	2.55438300	6.99617000	-0.68889300
H	4.28659200	5.23535700	-0.54196300
C	3.94377200	-2.35598100	-0.10709600
H	2.73599800	-1.22181600	-1.46612900
C	3.56075000	-1.19797800	-0.76042300
C	5.70544400	-1.08827500	1.01589700
C	4.12183500	0.05860000	-0.44290300
C	5.07978000	-2.29005300	0.75294300
C	5.20496400	0.13300600	0.48391800
C	3.48913400	1.24372400	-0.88176400
H	6.47622500	1.51158900	1.57565900
H	6.54792700	-1.05697000	1.70315400
C	3.80383100	2.47825800	-0.34480500
H	2.66276100	1.14914500	-1.57997300
C	4.94176200	2.55782800	0.51368300
C	5.63510700	1.42655500	0.89128700
H	-3.31687000	-4.35941800	-2.74489300
H	2.90276100	-6.94359700	-0.37174600
C	-5.39007000	0.36098200	-0.71790300
H	-5.50011200	1.14462400	-1.45965200
C	-5.98269700	-0.83303200	-0.79511100
H	-6.64296600	-1.14639300	-1.60024400
C	-5.65319400	-1.70172600	0.39437900
H	-6.50374100	-1.74333900	1.09131000
H	-5.41792400	-2.74048700	0.14296800
H	5.22791100	3.52203100	0.92522100
H	5.41996900	-3.19309800	1.25309400
O	-2.44392900	0.01564600	2.67263800
O	-0.90585700	0.01230400	0.90640500
C	0.04199900	-0.27557500	1.93274600
H	0.13973600	0.54885100	2.64814900
H	0.99626600	-0.40757800	1.41671700
H	-0.20146700	-1.18454500	2.49160900
C	-2.47159400	1.31972900	3.24529400
H	-3.35310300	1.88728600	2.92138300
H	-1.57641900	1.90540800	3.00028300
H	-2.51790300	1.17318500	4.32739600
C	0.73186200	-4.31952100	-0.44277900

TS3 charge 0, multiplicity 1, imaginary frequencies -480.24			
opt=(TS,nofreeze,readfc,noeig,MaxCycle=300,ModRedundant)			
freq ub3lyp/6-31g(d) iop(1/8=3) geom=allcheck			
guess=(mix,always) scfcyc=300 test			
Zero-point correction=		0.650071 (Hartree/Particle)	
Thermal correction to Energy=		0.686245	
Thermal correction to Enthalpy=		0.687190	
Thermal correction to Gibbs Free Energy=		0.583763	
Sum of electronic and zero-point Energies=		-1846.845876	
Sum of electronic and thermal Energies=		-1846.809701	
Sum of electronic and thermal Enthalpies=		-1846.808757	
Sum of electronic and thermal Free Energies=		-1846.912183	
C	2.25059000	-0.44431800	0.08073900
C	3.22962600	0.55189800	-0.19991600
C	2.80742300	-1.75725500	-0.09904300
C	4.27867300	-1.63510000	-0.43589400
H	4.51514300	-2.17376000	-1.36505400
C	4.53667500	-0.10162900	-0.57851900
H	4.80524800	0.15431100	-1.61025600
C	2.09856200	-3.01654400	-0.09463700
C	0.69922100	-5.47584600	-0.24309900
C	0.69580900	-3.08840500	0.09387800
C	2.78094200	-4.23689000	-0.33955600
C	2.08580300	-5.43767100	-0.40650200
C	-0.01736700	-4.29217700	0.00960700
H	0.17130700	-2.16676500	0.30456400
H	3.85437000	-4.23855200	-0.49659900
H	2.62702000	-6.35864800	-0.60769500
H	0.17350400	-6.42086200	-0.34080400
C	3.05524000	1.99836300	-0.19317600
C	2.67559700	4.79117100	-0.37370600
C	4.11302400	2.85364200	-0.58033900
C	1.81311000	2.59508900	0.11478200
C	1.59488900	3.97372100	0.00145000
C	3.92157700	4.22986600	-0.65259200
H	5.07941900	2.44360500	-0.85075500
H	1.00277400	1.94600600	0.41145200
H	4.74397500	4.87032600	-0.95966500
H	2.53117600	5.86194500	-0.48607500
C	-1.49761800	-4.29352300	0.18358900
C	-4.29429100	-4.15109800	0.59088700
C	-2.22599900	-3.13352800	-0.10147200
C	-2.21308800	-5.40788600	0.65561500
C	-3.59613400	-5.33688200	0.83149300
C	-3.60333300	-3.01671800	0.14249800
H	-1.71098100	-2.28554000	-0.53644300



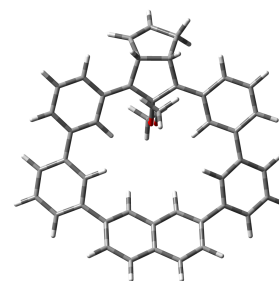
H	-1.69005400	-6.32489200	0.91155700
H	-4.13207800	-6.20845700	1.19853900
H	-5.36114000	-4.09727300	0.79159700
C	0.23160500	4.52530300	0.23303800
C	-2.40473800	5.43062600	0.69005700
C	-0.00454300	5.81965000	0.72598700
C	-0.87864000	3.72076500	-0.04616100
C	-2.19617200	4.12710600	0.21755600
C	-1.31134000	6.26654800	0.93001000
H	0.82932000	6.47144800	0.97239700
H	-0.71726800	2.75049900	-0.50212000
H	-1.47860700	7.26945400	1.31439900
H	-3.41173600	5.77778100	0.90648200
O	1.26108500	0.12368600	-1.53862600
O	1.14884400	-0.21362700	0.89187200
C	5.72598700	0.23740400	0.39057000
H	6.61160700	0.56978800	-0.16905400
H	5.47923100	1.04980500	1.08484100
C	5.21950300	-2.05891300	0.67899400
H	5.23371700	-3.07192200	1.06966400
C	5.98344200	-1.05790000	1.11599500
H	6.71705800	-1.13592500	1.91451300
C	-4.21037100	-1.67013900	-0.01459100
H	-2.46918700	-0.73911800	0.81629600
C	-3.44107600	-0.58438700	0.35853000
C	-5.94889900	-0.12132900	-0.76243200
C	-3.85080300	0.74326700	0.12244000
C	-5.50383800	-1.41274800	-0.55821100
C	-5.12721900	1.00124400	-0.45960600
C	-2.96852700	1.81050700	0.38547900
H	-6.13305100	-2.24955000	-0.85071400
H	-6.43308200	2.58397600	-1.16605400
H	-6.93068900	0.04753800	-1.19922100
C	-3.26884700	3.11484600	0.04218900
H	-2.00974500	1.57456600	0.83633400
C	-4.56216300	3.37961600	-0.49828800
H	-4.82654900	4.39909000	-0.76749800
C	-5.46214600	2.35814300	-0.73075600
C	1.45096700	-0.33814800	2.28590600
H	2.19727200	0.40370800	2.59876100
H	1.81651100	-1.34331600	2.52758200
H	0.51321300	-0.15347200	2.81417000
C	1.44349300	-0.71514200	-2.61446900
H	0.54969200	-1.34254900	-2.80305400
H	2.26465500	-1.44332500	-2.43356200
H	1.68088200	-0.16800900	-3.54446600

TS4 □charge □ 0, multiplicity□1, imaginary frequencies□ -486.96□			
opt=(TS,nofreeze,readfc,noeig,MaxCycle=300,ModRedundant)			
freq ub3lyp/6-31g(d) iop(1/8=3) geom=allcheck			
guess=(mix,always) scfcyc=300 test			
Zero-point correction=		0.649953 (Hartree/Particle)	
Thermal correction to Energy=		0.686140	
Thermal correction to Enthalpy=		0.687084	
Thermal correction to Gibbs Free Energy=		0.583538	
Sum of electronic and zero-point Energies=		-1846.848500	
Sum of electronic and thermal Energies=		-1846.812313	
Sum of electronic and thermal Enthalpies=		-1846.811369	
Sum of electronic and thermal Free Energies=		-1846.914915	
C	-2.20256400	-0.51975300	-0.26869600
C	-2.73765700	-1.83951500	-0.27411300
C	-3.26869500	0.43778400	-0.35874300
C	-4.60336100	-0.28400000	-0.38241200
H	-5.12198900	-0.08499900	-1.33492300
C	-4.23911900	-1.79490700	-0.25254600
H	-4.63787500	-2.35500200	-1.10724700
C	-3.12521800	1.87188700	-0.47590200
C	-2.81456600	4.67119400	-0.75692900
C	-1.90741800	2.51997800	-0.15728200
C	-4.20084400	2.68675600	-0.90978400
C	-4.03792200	4.06107200	-1.04368000
C	-1.72745600	3.90150900	-0.30450600
H	-1.10229900	1.91290600	0.23385400
H	-5.15647100	2.23639000	-1.16018500
H	-4.86967200	4.66726600	-1.39348300
H	-2.69859900	5.74053700	-0.90713600
C	-1.98029800	-3.07559700	-0.39869600
C	-0.48369300	-5.43803100	-0.73977800
C	-2.60217900	-4.26792200	-0.83390300
C	-0.60158300	-3.11908700	-0.09878900
C	0.16401600	-4.27918200	-0.27613900
C	-1.85471800	-5.42954600	-1.00102600
H	-3.66361300	-4.28260900	-1.05883500
H	-0.15756200	-2.23260400	0.32945700
H	-2.34045300	-6.33539300	-1.35354500
H	0.08847600	-6.34509200	-0.91314400
C	-0.40411400	4.51022300	0.00629900
C	2.15823900	5.52929300	0.63562500
C	0.75230000	3.73500000	-0.13250300
C	-0.24839800	5.83411100	0.45109900
C	1.02194200	6.33488600	0.74264400
C	2.03067700	4.19842200	0.21307300
H	0.66333100	2.73491400	-0.53916100



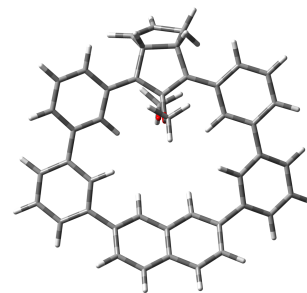
H	-1.11912000	6.46789100	0.59403800
H	1.12397300	7.35989500	1.08976700
H	3.13215300	5.92291300	0.91468800
C	1.62371300	-4.24833700	0.01499600
C	4.37918600	-4.03729800	0.60575100
C	2.35367100	-5.37352000	0.43342000
C	2.31461000	-3.03932900	-0.11840400
C	3.67133800	-2.89407800	0.20757000
C	3.71840600	-5.26394200	0.70709700
H	1.85524200	-6.32906500	0.57123800
H	1.78398500	-2.17703500	-0.50331700
H	4.26846300	-6.14207800	1.03534500
H	5.43086200	-3.96376300	0.86996800
O	-0.92017600	-0.23819800	-0.71674200
O	-1.58124900	-0.82084800	1.57651600
C	-4.91419800	-2.31186500	1.06292100
H	-5.55229900	-3.18485600	0.87365800
H	-4.16468800	-2.62363300	1.80128800
C	-5.53658400	-0.03649000	0.78664000
H	-5.98778900	0.93374700	0.97150400
C	-5.70419200	-1.12054200	1.54627300
H	-6.31292600	-1.16024900	2.44616900
C	3.14898000	3.22282600	0.14621500
H	1.91053000	1.65154800	0.91505700
C	2.88006800	1.91430800	0.50309800
C	5.40547500	2.53846000	-0.49959700
C	3.81030900	0.87432900	0.29327500
C	4.45834000	3.52844800	-0.32994800
C	5.10270400	1.17348400	-0.23300500
C	3.43017300	-0.46839600	0.50256400
H	4.69786900	4.55305700	-0.60281100
H	6.96806000	0.28270700	-0.89230600
H	6.38896100	2.79327700	-0.88824000
C	4.24561100	-1.52562700	0.14269100
H	2.44432300	-0.65711100	0.91669400
C	5.55551300	-1.22522100	-0.33555000
H	6.21959600	-2.04034700	-0.61158400
C	5.97282600	0.07995200	-0.50293300
C	-2.24586700	-0.04237300	2.50058400
H	-2.55179900	-0.61436700	3.39470300
H	-3.16131900	0.41726800	2.07112000
H	-1.62330900	0.81026700	2.83743700
C	-0.82522200	-0.16490100	-2.14329500
H	0.22228400	0.05292200	-2.36290800
H	-1.45638800	0.63539200	-2.54745400
H	-1.10191600	-1.11911200	-2.60994000

TS5 □ charge □ 0, multiplicity □ 3, imaginary frequencies □ -295.89 □			
opt=(TS,nofreeze,readfc,noeig,MaxCycle=300,ModRedundant)			
freq ub3lyp/6-31g(d) iop(1/8=3) geom=allcheck scfcyc=300 test			
Zero-point correction=		0.649632 (Hartree/Particle)	
Thermal correction to Energy=		0.686090	
Thermal correction to Enthalpy=		0.687034	
Thermal correction to Gibbs Free Energy=		0.581154	
Sum of electronic and zero-point Energies=		-1846.849185	
Sum of electronic and thermal Energies=		-1846.812726	
Sum of electronic and thermal Enthalpies=		-1846.811782	
Sum of electronic and thermal Free Energies=		-1846.917662	
C	-2.31556900	-0.28896400	-0.18832000
C	-3.21845200	0.79660800	0.07514600
C	-2.92387800	-1.56410600	0.06933000
C	-4.36163900	-1.32662100	0.48824300
H	-4.56803100	-1.79559400	1.46171600
C	-4.54313300	0.23072200	0.53709200
H	-4.76760800	0.56645500	1.55718500
C	-2.29520500	-2.86283000	0.10447100
C	-1.05499100	-5.39862600	0.34267200
C	-0.90922100	-3.03748100	-0.13292700
C	-3.04319600	-4.01814000	0.45179300
C	-2.42737400	-5.25940400	0.55724600
C	-0.27457200	-4.27917700	-0.00000800
H	-0.33255500	-2.16638300	-0.40926600
H	-4.10570500	-3.93871700	0.65440200
H	-3.01884700	-6.12885300	0.83247400
H	-0.58517500	-6.36872100	0.47524100
C	-2.91786100	2.21041000	0.11251800
C	-2.31813800	4.96855800	0.36054700
C	-3.91504000	3.15027500	0.48217000
C	-1.61820400	2.71419100	-0.14061100
C	-1.29745300	4.07080800	-0.00167200
C	-3.61341900	4.50262400	0.59160600
H	-4.92428200	2.81736400	0.69794400
H	-0.85220500	2.00942600	-0.43081300
H	-4.39200100	5.20271100	0.88359500
H	-2.09214700	6.02204900	0.49681300
C	1.19810200	-4.38447400	-0.19640900
C	3.99475900	-4.44548000	-0.60927100
C	2.00501700	-3.27345700	0.07146300
C	1.83018200	-5.55249300	-0.65661600
C	3.21403300	-5.58106400	-0.83804200
C	3.38761700	-3.26015800	-0.17124900
H	1.54781300	-2.38834000	0.49831600



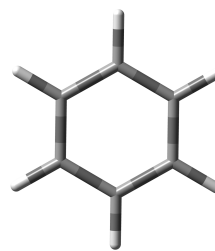
H	1.24123500	-6.43338200	-0.89651600
H	3.68636200	-6.49200700	-1.19682300
H	5.06288500	-4.47027500	-0.80898800
C	0.10556000	4.52613200	-0.20726100
C	2.80395300	5.25790100	-0.62758100
C	0.43648800	5.81153300	-0.66951000
C	1.15655100	3.64170200	0.05803800
C	2.50125000	3.96182400	-0.18702300
C	1.77230600	6.17184300	-0.85537400
H	-0.34807800	6.52468700	-0.90686400
H	0.92625100	2.67310600	0.48650600
H	2.01072000	7.16923300	-1.21595300
H	3.83428900	5.53917000	-0.82932700
O	-1.09221900	-0.14199300	1.43424000
O	-1.20663700	-0.15038500	-1.00035600
C	-5.75422100	0.56169700	-0.40503200
H	-6.60000000	0.98167400	0.15848400
H	-5.49108100	1.30825700	-1.16476200
C	-5.37787700	-1.77437600	-0.55036800
H	-5.46622200	-2.80940100	-0.86608000
C	-6.10985200	-0.76533600	-1.02217900
H	-6.89227400	-0.86459200	-1.77038400
C	4.09410000	-1.96359600	-0.01223700
H	2.43556700	-0.90259400	-0.85524200
C	3.41191500	-0.82213500	-0.38783900
C	5.93455600	-0.55277900	0.76397200
C	3.91525100	0.47001300	-0.13587800
C	5.39766700	-1.80627700	0.54530400
C	5.19978000	0.62909500	0.46381800
C	3.11446200	1.60081400	-0.39392700
H	5.95977400	-2.68937100	0.83838500
H	6.60670200	2.10553000	1.20537800
H	6.92125300	-0.45967000	1.21230800
C	3.50031000	2.87514500	-0.02445700
H	2.14758200	1.44038200	-0.86044200
C	4.80308300	3.04040200	0.53270400
H	5.13484900	4.03464700	0.82099000
C	5.62699500	1.95504300	0.75730200
C	-1.53121500	-0.19287100	-2.39380100
H	-2.19666200	0.63459300	-2.66980300
H	-2.00423600	-1.14581100	-2.66104000
H	-0.58386300	-0.09252300	-2.92734500
C	-1.67031200	-0.21360000	2.69290500
H	-0.84193500	-0.11895100	3.41633000
H	-2.16572100	-1.17873600	2.89608100
H	-2.37448400	0.61030600	2.90193600

TS6 □charge □ 0, multiplicity□3, imaginary frequencies□ -318.08□			
opt=(TS,nofreeze,readfc,noeig,MaxCycle=300,ModRedundant)			
freq ub3lyp/6-31g(d) iop(1/8=3) geom=allcheck scfcyc=300 test			
Zero-point correction=		0.649345 (Hartree/Particle)	
Thermal correction to Energy=		0.685860	
Thermal correction to Enthalpy=		0.686804	
Thermal correction to Gibbs Free Energy=		0.580349	
Sum of electronic and zero-point Energies=		-1846.849586	
Sum of electronic and thermal Energies=		-1846.813071	
Sum of electronic and thermal Enthalpies=		-1846.812127	
Sum of electronic and thermal Free Energies=		-1846.918582	
C	1.91863700	-1.23676400	-0.37938200
C	3.24255700	-0.67710800	-0.37720700
C	1.96235500	-2.67139300	-0.35874600
C	3.41098700	-3.11294500	-0.27684100
H	3.64378500	-3.78187900	-1.11988700
C	4.26068900	-1.79549800	-0.33750700
H	4.86100200	-1.77019900	-1.25671900
C	0.84133300	-3.57851900	-0.43719200
C	-1.36875600	-5.32630400	-0.66634400
C	-0.47439300	-3.14972100	-0.14032400
C	1.01984000	-4.93525300	-0.80707800
C	-0.07430600	-5.78556900	-0.92003800
C	-1.58469800	-3.99575600	-0.26304500
H	-0.59692100	-2.14359000	0.23570200
H	2.01332800	-5.31311100	-1.02832000
H	0.07805500	-6.81753400	-1.22570700
H	-2.21190200	-5.99792800	-0.79888600
C	3.58645300	0.72303500	-0.46112100
C	4.22895300	3.46522000	-0.69971000
C	4.88738300	1.13635600	-0.84403600
C	2.64221200	1.73283900	-0.15649900
C	2.93625200	3.09703100	-0.28378800
C	5.19178500	2.48761500	-0.96189000
H	5.64831400	0.39568500	-1.07017500
H	1.68121700	1.41660000	0.22551600
H	6.18779400	2.78745800	-1.27755700
H	4.47796500	4.51354100	-0.83603600
C	-2.94777700	-3.47299200	0.03150600
C	-5.47121200	-2.34884300	0.63944100
C	-3.20346700	-2.10662900	-0.12662800
C	-4.00387300	-4.28262800	0.48377600
C	-5.25190400	-3.72286000	0.76349600
C	-4.42980100	-1.51391800	0.21008500
H	-2.42498100	-1.47568200	-0.53798600
H	-3.84778200	-5.34594800	0.64239500

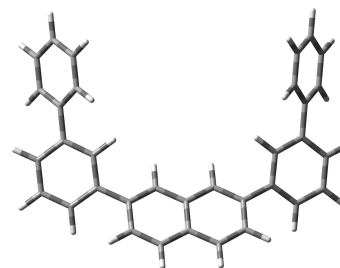


H	-6.05644700	-4.36327100	1.11590300
H	-6.43439200	-1.92507800	0.91156700
C	1.88688600	4.11022300	0.01642400
C	-0.19653000	5.92167900	0.63048700
C	2.17572500	5.40992700	0.46661700
C	0.54008900	3.76328500	-0.13580400
C	-0.51439400	4.62458500	0.20318800
C	1.14168700	6.30415900	0.74983600
H	3.20597000	5.71833900	0.62054100
H	0.29641100	2.79040700	-0.54517300
H	1.38293000	7.30431600	1.10067800
H	-0.98695000	6.61562000	0.90435900
O	0.83892900	-0.55144200	-0.90338000
O	1.19196700	-0.76959500	1.45423000
C	5.21529200	-1.82882300	0.90160900
H	6.26922700	-1.88920300	0.59552800
H	5.12572700	-0.92244600	1.51333700
C	3.83278600	-3.75599800	1.03084100
H	3.37792900	-4.67260500	1.39479700
C	4.79146100	-3.06720000	1.65258200
H	5.23945600	-3.34845700	2.60250300
C	-4.51402500	-0.03296500	0.13375800
H	-2.51996000	0.18250000	0.89040900
C	-3.38884600	0.69052400	0.48323800
C	-5.60756500	2.05789100	-0.50710400
C	-3.29891500	2.08267200	0.27384500
C	-5.65074600	0.68842500	-0.33756700
C	-4.42005000	2.79736800	-0.24451800
C	-2.07395200	2.75109200	0.48081100
H	-6.55170700	0.14254100	-0.60581300
H	-5.09377200	4.75545800	-0.89249600
H	-6.48001300	2.58198600	-0.89106200
C	-1.89275100	4.07604300	0.12985800
H	-1.24636300	2.17740300	0.88687800
C	-3.02664300	4.80232600	-0.34082700
H	-2.91170700	5.84917200	-0.61022800
C	-4.25039600	4.18566900	-0.50867200
C	1.94048000	-0.98642700	2.60385900
H	2.87327600	-0.39899300	2.63884500
H	2.17334100	-2.04694000	2.78268400
H	1.30897500	-0.64476200	3.44297100
C	0.81054700	-0.55368400	-2.33355600
H	-0.07686300	0.01643600	-2.61620300
H	0.72918100	-1.57379700	-2.72862200
H	1.70238000	-0.06873000	-2.74953800

Benzene □charge □ 0, multiplicity□1□			
opt freq rb3lyp/6-31g(d)			
Zero-point correction=	0.100754 (Hartree/Particle)		
Thermal correction to Energy=	0.105143		
Thermal correction to Enthalpy=	0.106087		
Thermal correction to Gibbs Free Energy=	0.073292		
Sum of electronic and zero-point Energies=	-232.147897		
Sum of electronic and thermal Energies=	-232.143508		
Sum of electronic and thermal Enthalpies=	-232.142564		
Sum of electronic and thermal Free Energies=	-232.175359		
C	-1.33923600	-0.39585300	0.00000500
C	-0.32668500	-1.35773000	0.00006900
C	1.01242100	-0.96187800	-0.00005500
C	1.33920200	0.39596500	0.00000800
C	0.32679800	1.35770000	0.00006200
C	-1.01250100	0.96179800	-0.00005800
H	-2.38133000	-0.70401000	-0.00006600
H	-0.58108600	-2.41424500	0.00007900
H	1.80037500	-1.71023400	-0.00015400
H	2.38136700	0.70385800	0.00000700
H	0.58095200	2.41426700	0.00001800
H	-1.80027600	1.71035300	-0.00006300

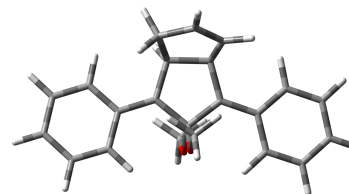


Optimized macrocyclic skeleton in π-3b □charge □ 0, multiplicity□1□			
opt freq rb3lyp/6-31g(d)			
Zero-point correction=	0.471810 (Hartree/Particle)		
Thermal correction to Energy=	0.497442		
Thermal correction to Enthalpy=	0.498386		
Thermal correction to Gibbs Free Energy=	0.413171		
Sum of electronic and zero-point Energies=	-1309.650301		
Sum of electronic and thermal Energies=	-1309.624669		
Sum of electronic and thermal Enthalpies=	-1309.623725		
Sum of electronic and thermal Free Energies=	-1309.708940		
C	-4.79909100	3.88218900	1.28787800
C	-5.96656300	2.70449500	-0.94941600
C	-4.66671700	2.51130200	1.07144900
C	-5.51662700	4.67071800	0.38659300
C	-6.10014100	4.07526800	-0.73316300
C	-5.24744400	1.89776900	-0.05137300
H	-4.12915800	1.90294800	1.79353600
H	-5.62022000	5.73918200	0.55534700
H	-6.40229900	2.25573200	-1.83785800
C	4.79910400	3.88217900	1.28792100
C	5.96658000	2.70450300	-0.94937700
C	4.66672400	2.51128900	1.07147200



C	5.51664100	4.67071300	0.38665500
C	6.10015600	4.07526800	-0.73310900
C	5.24745200	1.89775900	-0.05134600
H	4.12915600	1.90294400	1.79355900
H	5.62023400	5.73917700	0.55540700
H	6.65408400	4.68057600	-1.44599200
H	6.40230900	2.25575100	-1.83782800
C	-4.83432900	-2.33161600	-0.70222500
C	-3.90259400	-0.22290900	0.00225400
C	-6.17734000	-0.32052800	-0.78821800
C	-6.03753800	-1.69171200	-0.99332500
C	-3.74364100	-1.60190100	-0.20083700
H	-3.06748700	0.35054100	0.39353700
H	-7.12850000	0.16303300	-0.99144400
H	-4.72945700	-3.39636400	-0.88984000
C	5.10667800	0.43662600	-0.28458600
C	4.83429900	-2.33162100	-0.70229800
C	6.17731800	-0.32053700	-0.78826900
C	3.90259800	-0.22292100	0.00226600
C	3.74363500	-1.60190500	-0.20087400
C	6.03750700	-1.69171500	-0.99340500
H	7.12848100	0.16301700	-0.99149600
H	3.06749200	0.35051100	0.39357600
H	6.87378200	-2.26692200	-1.38214100
H	4.72941000	-3.39636700	-0.88992100
C	-2.45091400	-2.27060100	0.09753100
H	-1.22865600	-0.67063600	-0.63905000
C	-1.24124300	-1.65308000	-0.17371400
C	-1.24701000	-4.21226900	0.95586000
C	-0.00000200	-2.28065100	0.10889900
C	-2.43191800	-3.57575600	0.67249900
C	0.00000600	-3.59155700	0.68335800
C	1.24123700	-1.65307400	-0.17372800
H	1.25323500	-5.20335300	1.40378800
H	-1.25321600	-5.20334400	1.40381900
C	2.45090500	-2.27059600	0.09750800
H	1.22865100	-0.67062400	-0.63905300
C	2.43192700	-3.57575700	0.67246500
C	1.24702400	-4.21227300	0.95584200
H	-6.65407800	4.68056700	-1.44604600
H	-6.87381200	-2.26693000	-1.38204600
H	3.37443900	-4.05930500	0.91243200
H	-3.37441800	-4.05930500	0.91251100
C	-5.10667600	0.43663100	-0.28458600
H	-4.34788200	4.33341900	2.16777200
H	4.34789900	4.33337200	2.16783600

Standard structure in π -3b [charge = 0, multiplicity=1]			
opt freq ub3lyp/6-31g(d) guess=(mix,always)			
Zero-point correction=	0.382671 (Hartree/Particle)		
Thermal correction to Energy=	0.403818		
Thermal correction to Enthalpy=	0.404762		
Thermal correction to Gibbs Free Energy=	0.332873		
Sum of electronic and zero-point Energies=	-1001.535990		
Sum of electronic and thermal Energies=	-1001.514843		
Sum of electronic and thermal Enthalpies=	-1001.513898		
Sum of electronic and thermal Free Energies=	-1001.585788		
C	0.78109700	1.64958700	-0.69190200
H	1.21089000	1.87262200	-1.68226100
C	1.20943400	0.28216800	-0.22366400
C	0.02014000	-0.59067800	0.12369300
C	-1.17922200	0.27168200	-0.22147100
C	-0.78082600	1.62507600	-0.74294800
H	-1.12643100	1.75050600	-1.77882600
C	-1.26978600	2.84796000	0.11179600
H	-1.77977300	3.59503500	-0.51286500
H	-1.99138100	2.54789900	0.88208300
C	-0.00263100	3.40557000	0.70698600
H	-0.00831200	4.23311200	1.41249100
C	1.09355300	2.77886900	0.27803300
H	2.10963500	3.01190800	0.58217000
O	0.02615600	-1.85308900	-0.55778600
O	0.02695800	-1.07706100	1.47373400
C	0.02174500	-1.79518900	-1.97312400
H	0.03167100	-2.83337300	-2.31407100
H	0.90976900	-1.28354200	-2.36962700
H	-0.87948700	-1.30279300	-2.36430800
C	-0.00867100	-0.09182900	2.49213800
H	-0.94018000	0.49076000	2.46878600
H	0.84111600	0.60041900	2.42954100
H	0.04344000	-0.63860300	3.43688200
C	2.56994000	-0.15003500	-0.15272600
C	5.27917500	-0.98596100	-0.02493500
C	2.92782400	-1.42521200	0.37478300
C	3.62936900	0.68231500	-0.61386100
C	4.95354800	0.26928500	-0.55012300
C	4.25526900	-1.82443700	0.43398400
H	2.14498700	-2.07992300	0.73553000
H	3.40100500	1.65625600	-1.03504500
H	5.73888100	0.92724400	-0.91365800
H	6.31602800	-1.30690600	0.02541100
C	-2.53349200	-0.18447100	-0.14473500



C	-5.22667600	-1.06993200	-0.01278700
C	-2.87094900	-1.44827900	0.42114300
C	-3.60448500	0.61063500	-0.64243300
C	-4.92092100	0.17343700	-0.57660000
C	-4.19086400	-1.87223600	0.48207700
H	-2.07859500	-2.07538900	0.80932900
H	-3.39076200	1.57417300	-1.09397100
H	-5.71558500	0.80290200	-0.96918000
H	-6.25745000	-1.40972100	0.03911500
H	4.49966900	-2.80124500	0.84377800
H	-4.41947800	-2.84008300	0.92117700

Standard structure in σ -3b [charge = 0, multiplicity=1]

opt freq rb3lyp/6-31g(d)

Zero-point correction= 0.384877 (Hartree/Particle)

Thermal correction to Energy= 0.405754

Thermal correction to Enthalpy= 0.406698

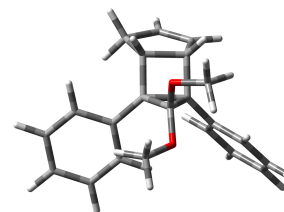
Thermal correction to Gibbs Free Energy= 0.335442

Sum of electronic and zero-point Energies= -1001.550260

Sum of electronic and thermal Energies= -1001.529383

Sum of electronic and thermal Enthalpies= -1001.528439

Sum of electronic and thermal Free Energies= -1001.599695



C	-0.79393800	-1.76094900	1.22127000
H	-1.23671200	-2.70235700	0.88815700
C	-0.77623500	-0.66311600	0.13467700
C	0.06416800	-1.16271700	-1.03060400
C	0.80704200	-0.62890900	0.15844100
C	0.77551100	-1.71967800	1.26669300
H	1.31358600	-2.63852800	1.00301200
C	-1.79772700	0.40836800	-0.01464100
C	-3.80350600	2.37055100	-0.31508100
C	-3.14673400	0.11255700	0.25164700
C	-1.47439400	1.70927300	-0.43477900
C	-2.46845200	2.67691400	-0.58386500
C	-4.13882600	1.08144200	0.10282300
H	-3.42041100	-0.89018600	0.56955600
H	-0.44227100	1.95953100	-0.65059400
H	-4.57451000	3.12795100	-0.42919900
C	1.79476300	0.48745400	0.10588700
C	3.73830700	2.52777600	0.07788600
C	2.81177200	0.49972900	-0.86157700
C	1.76691900	1.52138300	1.05647100
C	2.72949800	2.53131200	1.04288200
C	3.77476100	1.50866100	-0.87552100
H	2.83770300	-0.28145000	-1.61477300

H	0.98398900	1.52922400	1.80807700
H	2.68858200	3.32363400	1.78607200
H	4.48734800	3.31531400	0.06615400
H	-2.19482700	3.67743000	-0.90949000
C	1.09118700	-1.24799600	2.65995900
H	2.10916500	-1.08546900	3.00257800
C	-0.00709900	-1.05829300	3.39556100
H	-0.01147900	-0.72699500	4.43124500
C	-1.28053100	-1.35183900	2.63753700
H	-1.85339600	-2.16086700	3.11331600
H	-1.94825100	-0.48114900	2.61442800
O	0.01231100	-2.52543900	-1.33261500
O	0.14544000	-0.34309300	-2.15145500
C	-0.92339300	-0.51674300	-3.08717200
H	-0.91447800	-1.52507200	-3.51542500
H	-0.75263200	0.21933600	-3.87565300
H	-1.89634600	-0.33232400	-2.61974700
C	1.20207100	-3.02455100	-1.93977100
H	1.05200400	-4.09795500	-2.07778700
H	2.07660500	-2.85975600	-1.29567300
H	1.38515000	-2.55308900	-2.91276600
H	-5.17460200	0.82716800	0.31348200
H	4.55278100	1.50009400	-1.63475400
